# Integrating societal preferences: Decentralization of the European energy network

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**Abstract**

Standard EU energy system modelling approaches optimize for least-cost, leading to highly centralized systems, in conflict with political feasibility and physical security concerns. This master's thesis begins to bridge the gap between techno-economic energy system models, and societal paradigms by incorporating decentralization as a constraint in a European energy system model. The model is a 37-node electricity-only brownfield system based on the PyPSA-EUR framework, with projected 2050 loads and technology costs. A total of 105 optimized scenarios are analyzed at 14 levels of decarbonization (0%-100% $CO_2$ reduction) and 8 levels of decentralization. The novelty of this work lies in the systematic exploration of interactions between decarbonization and decentralization using a load weighted renewable capacity constraint (K-parameter). The results show that full decarbonization leads to an 80% cost increase due to, among other factors, a 78% increase in energy generation capacity. It is found that the system behavior changes sharply at around 80%-90% decarbonization, where resource quality becomes critical. As for decentralization, it is shown that equity improves naturally when decarbonizing up until 80% where the system begins to re-centralize aggressively. It is found that complete decentralization as enforced by K=1 leads to a 18% cost increase at full decarbonization compared to the least-cost scenario. However moderate decentralization of K=7 achieves 76% of the equity benefits at only a 9% cost increase compared to K=1. This indicates that moderate decentralization can be a viable strategy to balance societal preferences and cost-efficiency in the European energy transition.

*Keywords:* European energy system, Decentralization, Decarbonization, Paradigms in energy transitions, Renewable Energy Engineering

## Contents

## 1. Introduction

In a world where the necessity of reducing the carbon emissions of our energy networks is no longer a suggestion, but a requirement, careful and deliberate planning of the energy system of the future is needed. When the Paris Climate Agreement was signed into action in 2015, it signaled the commitment of the United Nations to reducing their overall carbon footprint [1]. This common goal and understanding of the necessity of decarbonization was a crucial step towards a more sustainable world, and provided a common understanding of what needs to be done. It also provides a massively complicated technological challenge that must be solved in order for the agreement to be fulfilled.

Somewhat unique to the issue of decarbonization in the European Union is the fact that, although the Paris Climate Agreement signifies the common goal of decarbonization, the EU is still comprised of many different countries with many different political ideologies, laws, societal preferences, and goals of their own. This means that, although it has long been known that a shared and interconnected energy grid is the most efficient and economically preferable solution, as seen, for example, in paper [2], reality becomes a lot more complicated as the member nations also must have their own best interests in mind.

Historically, the impact of the union being made up of self-centered actors has often been discarded, when trying to construct the electricity grid of the future, if for no other reason than the fact that implementing social preferences and societal paradigms in strict mathematical models of the European energy system is a challenging problem [3]. These models, that typically aim to reduce the overall cost of the system as the primary objective, are often meant to be used as the basis for developing the future energy network and legislation. However, using this least-cost optimization approach without considering social paradigms in the European Union tends to produce system models that, although being economically efficient at achieving decarbonization targets, are intrinsically tied to the assumption that the EU has complete control over the member nations' individual energy grids.

This approach of reducing the optimization problems to purely techno-economic problems that prioritize cost reduction over feasibility, has helped achieve significant insights into decarbonization strategies for the European Union [4], but tends to lead to the energy system becoming heavily centralized [5].

Centralization, in the context of energy systems, refers to the majority of resources being pooled into a few areas that are favorable to the system in some way. This could be installing a lot of wind turbines in a region with a lot of wind, maintaining energy production in regions close to where the energy is needed, or any number of other mechanisms that result in a proportionally large investment into infrastructure in a single region. While centralization as a strategy is often the most cost-efficient, it leads to a suite of other challenges.

Two such challenges are those of physical security and political feasibility. With the recent surge in so-called hybrid attacks on European energy infrastructure [6], the focus on physical security has increased. If a system is overly centralized, it means that a large amount of critical infrastructure is installed in a relatively small area. This makes the infrastructure an obvious target for such hybrid attacks, as a single hostile action can impact more infrastructure and more people. As for political feasibility, nations who end up as net energy importers become less self-sufficient and increasingly reliant on the net exporting regions. This shift in power balance heavily favors the net exporters and, as such, may face resistance from net importing nations that wish to remain at a high degree of self-sufficiency.

The objective of this master's thesis is to incorporate societal paradigms and priorities in the form of decentralization into a macro-scale energy network model of the EU, and to analyze and expose the mechanisms behind decarbonization and its interactions with decentralization. The main research questions are then:

- How, if at all, does enforcing decentralization change the cost-optimal pathway to decarbonization in a European energy system?
- Which system mechanisms are responsible for the interactions between decarbonization and decentralization?
- What sacrifices must be made in terms of cost, and system design, in order to achieve a decentralized energy system?

Answering these questions may then help develop a workflow that, by incorporating societal paradigms and reviewing their system impacts, will help close the gap between the models used to inform energy policy-makers, and the results of the proposed policies. Although this workflow and methodology may in time prove relevant for policy makers to understand, the target audience of this master's thesis remains peers in the field of energy system modeling who may draw inspiration from the intersection between societal paradigms and strict mathematical modeling that this work represents. It is also for this reason that all the data and code used to produce the results, which can be seen in this paper, are openly available in a public repository under an open science license[7].

## 2. Nomenclature

*Symbols*

| Symbol | Description |
|---|---|
| $n$ | Node index (countries/regions/nodes) |
| $t$ | Time snapshot index (hourly timesteps) |
| $g$ | Generator technology index |
| $C$ | Decarbonization level (fraction of baseline emissions reduction) |
| $K$ | Decentralization parameter (capacity distribution constraint factor) |
| $\lambda_{n,t}$ | Marginal electricity price (shadow price) at node $n$ and time $t$ (EUR/MWh) |
| $\mu_C$ | Shadow price (dual variable) of $CO_2$ emissions constraint at decarbonization level $C$ (EUR/t$CO_2$) |
| $p_{\text{nom}}$ | Nominal installed capacity (MW) |
| $p_{\text{existing}}$ | Pre-existing (brownfield) capacity (MW) |
| $p_{\text{line}}$ | Optimized transmission line capacity (MW) |
| $p_{\text{max\_pu},g,n,t}$ | Maximum power output availability factor for generator $g$ at node $n$ and time $t$ (per unit) |
| $p_{\text{min\_pu},g,n,t}$ | Minimum stable generation for generator $g$ at node $n$ and time $t$ (per unit) |
| $e_g$ | Emission factor for generator technology $g$ (t$CO_2$/$\text{MWh}_{\text{fuel}}$) |
| $w_t$ | Temporal weighting factor for snapshot $t$ |
| $L_n$ or $L_{n,t}$ | Electrical load (demand) at node $n$ (and time $t$) (MW or MWh) |
| $\omega_n$ | Normalized load-share weight for node $n$ ($= L_n/\sum_{n'} L_{n'}$) |
| $R_{\text{cap},n}$ | Installed renewable generation capacity at node $n$ (MW) |
| $R_{\text{total}}$ | Total renewable generation capacity across all nodes (MW) |
| $r$ | Discount rate (per annum) |
| $G$ | Gini coefficient of inequality |
| $\rho$ | Spatial correlation coefficient between renewable capacity and load |
| $\sigma$ | Standard deviation |
| $C_n^R$ | Renewable capacity at node $n$ (MW) |
| $\langle\cdot\rangle$ | Mean value operator across all nodes |

*Abbreviations*

| **Abbreviation** | **Definition** |
|---|---|
| AC | Alternating Current |
| CAPEX | Capital Expenditure |
| CCGT | Combined Cycle Gas Turbine |
| $CO_2$ | Carbon Dioxide |
| CV | Coefficient of Variation |
| DC | Direct Current |
| DEA | Danish Energy Agency |
| EGS | Enhanced Geothermal Systems |
| EU | European Union |
| FOM | Fixed Operation and Maintenance |
| HPC | High Performance Computing |
| HVDC | High Voltage Direct Current |
| KKT | Karush-Kuhn-Tucker |
| LCOE | Levelized Cost of Electricity |
| NEA | Nuclear Energy Agency |
| OCGT | Open Cycle Gas Turbine |
| PHS | Pumped Hydro Storage |
| PV | Photovoltaic |
| PyPSA | Python for Power System Analysis |
| SDRCL | Standard Deviation of Nodal Renewable Capacity per Load |

## 3. Methodology

### *3.1. Modeling framework*

For the purpose of structuring the modeling of the EU electricity system, a Snakemake workflow was constructed. Snakemake is a workflow management system that allows for modular and reproducible data analysis [8]. The workflow is divided into multiple steps or "rules" that, when chained together, ensure that the data is processed in the correct order. A workflow management system was implemented for two reasons. As documented later on in the thesis, the large number of scenarios simulated with the workflow lends itself well to a standardized workflow, that can handle small variations to construct different scenarios. Secondarily, Snakemake is already used in many power system optimization workflows and was therefore the obvious candidate. The complete workflow used to produce the results in this master's thesis is openly available in a public repository [9] under a GNU General Public License [10].

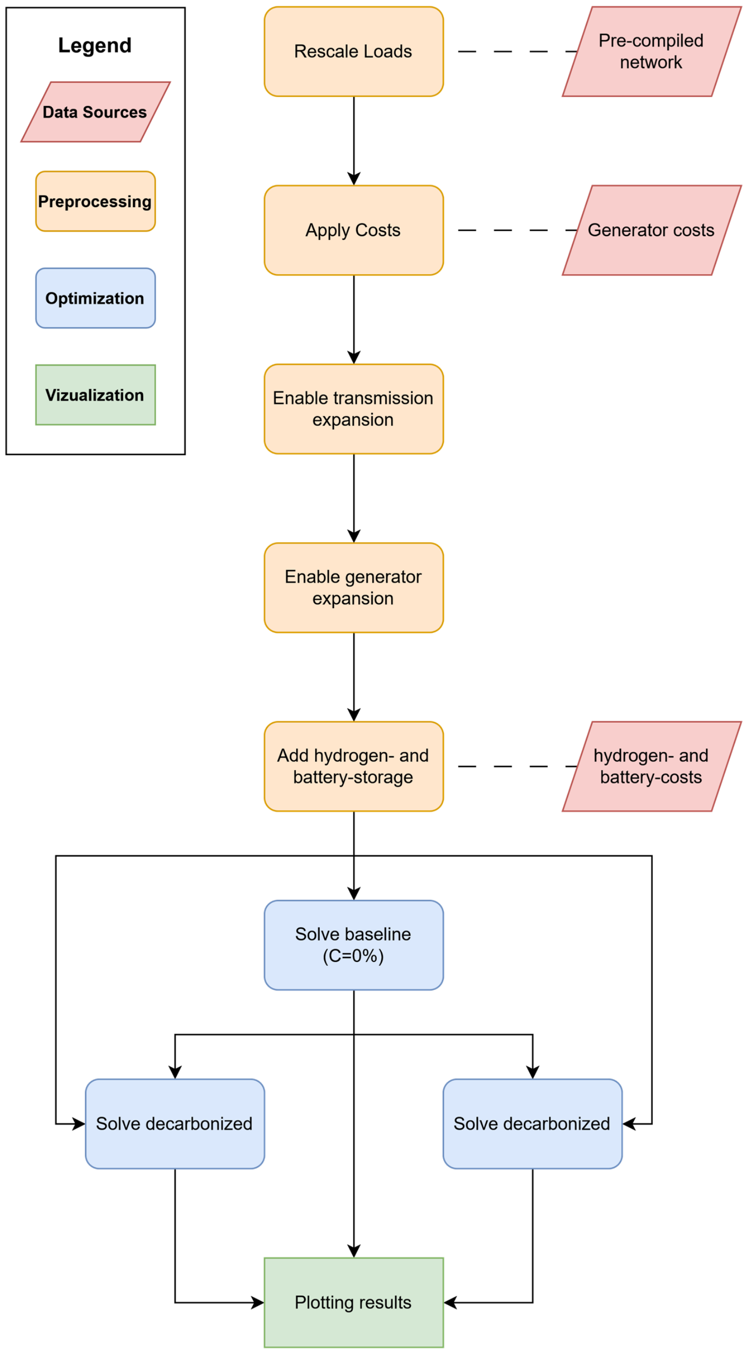


Figure 1: Map of Snakemake Workflow

Seen above in Figure 1 is a map of the rules and data input used in the workflow, divided into three main sections: preprocessing, optimization, and visualization/postprocessing. The following sections will go further into detail on the different sections of the workflow, excluding the visualizations and postprocessing, which are primarily used for producing figures and tables for this master's thesis results section.

### *3.2. Precompiled network model*

Many energy modeling frameworks and pieces of commercially available software for optimizing nodal networks exist. The work represented by this paper utilizes the open-source framework PyPSA (Python for Power System Analysis) [11]. The reasoning for this is twofold: Firstly, PyPSA is completely free and open source to use. This encourages open and collaborative science, as well as transparency in the philosophy behind the energy system modeling. Secondly, PyPSA consists of two separate parts: a modeling workflow based in the Python programming language, and a large database, containing relevant open-source data for constructing models, particularly of the European Union. This dataset is referred to as PyPSA-Eur[12] and is invaluable in its quality of data for building macro energy system models in the PyPSA workflow. This master's thesis utilizes both parts of the PyPSA environment. Acting as a baseline for all of the network topology for the European Union is a pre-compiled network developed with PyPSA-EUR using the default settings as published in the Zenodo repository [13].

The simple philosophy behind the PyPSA models is that a network is perceived as multiple interconnected nodes. These nodes and connections are then given a set of constraints that define the feasible solution space. Examples of constraints could be, that each node is assigned an electrical load that must be serviced based on historical consumption, capacity factor time series based on solar and wind patterns, or maximum or minimum capacity constraints, like a maximum allowable power flow from one node to another representing the transmission infrastructure in the network. The network used in this master's thesis is a 37-node network representing the 25 (excluding the island states of Malta and Cyprus) member nations of the EU. Additionally, Switzerland, Norway, Bosnia-Herzegovina, Albania, Montenegro, North Macedonia, the United Kingdom, and Serbia are included as their electrical infrastructure is connected to that of the EU. The following nations are also represented by more than one node as they have large populations on islands: Denmark, Spain, Great Britain, and Italy.

These nodes are then interconnected with HVDC and AC links and lines based on real-world connections. A graphical overview of the network can be seen below in Figure 2:

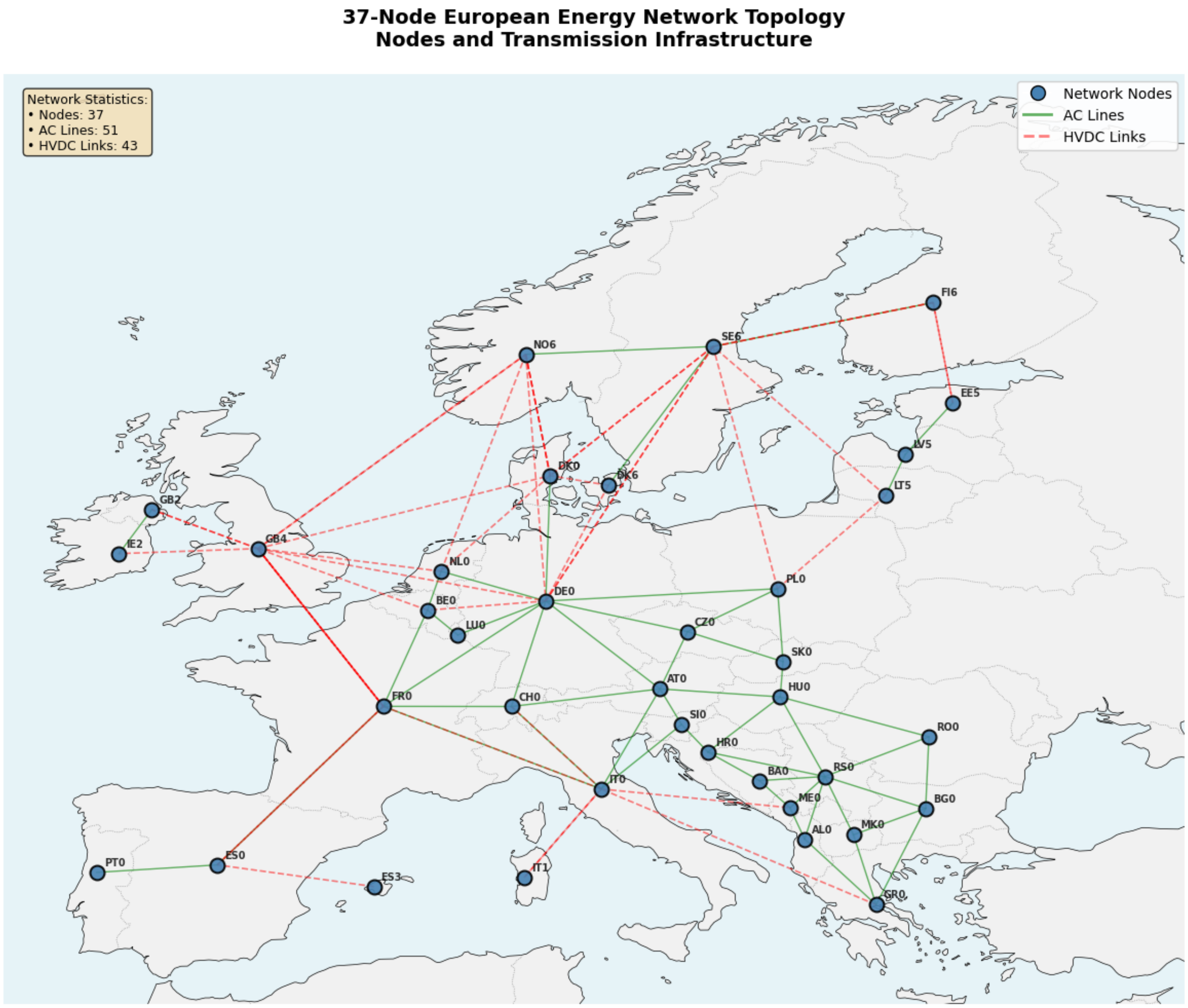


Figure 2: Overview of the 37-node European energy network.

The 37-node network is in actuality a clustered version of the standard PyPSA-EUR network, which contains 1,024 nodes. This is incredibly computationally intensive to solve. The effects of the clustering process on the network have been analyzed in [14], where it is found that similar power system models show relatively small sensitivity to increases and decreases in the number of nodes. In addition, it was found that a time-scale of 1-hour resolution was sufficient to capture the majority of time-dependent effects in the system.

### 3.3. Load rescaling

The precompiled network models come preloaded with electrical demand time series for each of the nodes from the ENTSOE-E transparency platform [15]. These hourly demand time series are, however, in the case of the precompiled model, from the year 2013. In order to ensure a more representative demand profile for the nodes, the loads have been rescaled to projected values for 2050. Each node is assigned a target electricity consumption value based on figures from Carbon-Free Europe [16]. These are then used together with the load values from 2013 to calculate a nodal scaling factor for each node, as seen in equation (1) below:

$$\text{scaling factor}_n = \frac{\text{target Load}_n}{\text{2013 Load}_n} \quad \forall\, n \tag{1}$$

Here $n$ denotes the set of all nodes. These nodal scaling factors are then used as weights on the hourly load time series for each node, to produce a rescaled load time series for each node as seen in equation (2) below:

$$\text{rescaled Load}_{n,t} = \text{scaling factor}_n \cdot \text{2013 Load}_{n,t} \quad \forall\, n, t \tag{2}$$

Where $t$ denotes the set of all time snapshots. This way the total load is scaled to the projected value but the general shape of the load time series is preserved. An example of a nodal load time series before and after rescaling can be seen below in Figure 3:

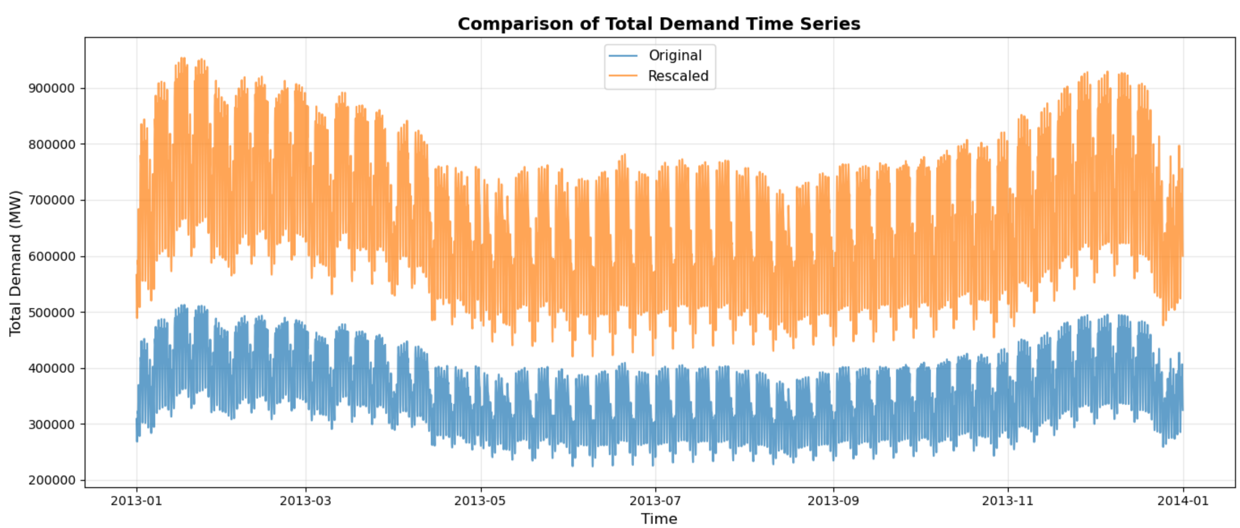


Figure 3: Example of load rescaling for node representing Denmark.

Here it can be seen that the shape of the load time series has been preserved while the total load has been rescaled to match the projected values. This rescaling process assumes that while the total energy consumption will increase the pattern will remain constant. This is of course a simplification as continuing electrification of different sectors, such as heating and transport, as well as changing weather patterns due to climate change, may very well alter the consumption pattern in the future. However, sector coupling and synthetic climate change models are beyond the scope of this master's thesis and this simplification is therefore necessary.

### *3.4. Generator Cost Modeling*

Although the precompiled network comes with some cost data already defined, there are some technologies that do not have cost data implemented. In order to be able to optimize the network fully this must be rectified. Table 1 below shows the cost data used for the different generation technologies in the network.

Table 1: Cost assumptions for generation technologies [17] [18] [19]

| **Carrier** | **CAPEX (EUR/MW)** | **Lifetime (years)** | **Discount rate** | **FOM (abs) (EUR/MW/a)** | **FOM (rate) (per a)** | **Notes** |
|---|---|---|---|---|---|---|
| CCGT | 830,000 | 25 | 0.07 | 27,800 | – | DEA 2030 |
| OCGT | 435,000 | 25 | 0.07 | 7,745 | – | DEA 2030 |
| Coal | 1,860,000 | 30 | 0.07 | 30,355 | – | DEA 2030 |
| Lignite | 1,860,000 | 30 | 0.07 | 30,355 | – | Proxied as coal CAPEX |
| Oil | 343,000 | 25 | 0.07 | 8,448 | – | DEA 2030 |
| Biomass | 3,300,000 | 25 | 0.07 | 96,000 | – | DEA 2030 |
| Geothermal | 4,290,000 | 20 | 0.07 | – | 0.02 | EGS ∼2% FOM |
| Nuclear | 5,870,000 | 60 | 0.07 | – | 0.02 | NEA 2020 ∼2% FOM |

PyPSA, stores generator cost parameters as annualized total ownership costs. Due to differences in the way different sources report costs, some conversion is necessary in order to convert the reported costs into the format used by PyPSA. Firstly, for sources that report the FOM (Fixed Operation and Maintenance) costs as an absolute value the total annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{FOM (abs)} \tag{3}$$

Where the Annuity Factor is calculated as:

$$\text{Annuity Factor} = \frac{r(1+r)^y}{(1+r)^y - 1} \tag{4}$$

With $r$ being the discount rate and $y$ being the lifetime of the technology in years. For sources that report FOM as an annual rate the annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{CAPEX} \cdot \text{FOM (rate)} \tag{5}$$

And finally for sources that report neither FOM (abs) nor FOM (rate) the annualized cost is simply calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} \tag{6}$$

Performing these operations ensures that all the cost data is formatted the same when interfacing with the PyPSA model. The full cost assumptions for the different types of generators can be found in Appendix A: Generator cost assumptions. It is also important to note that annualization is in fact a specific case of scaling costs over time, and the yearly cost is simply utilized because the optimization timespan included in the pre-compiled model is a single year.

### *3.5. Transmission- and generator-expansion*

In order to find the optimal network the transmission infrastructure as well as select generator types must be allowed to vary their capacity. For transmission infrastructure the line and link capacities was constrained as seen in equation (7) below:

$$p_{\text{existing}} \leq p_{\text{line}} \leq p_{\text{line, max}} \tag{7}$$

Where $p_{\text{existing}}$ is the existing capacity of the line or link, $p_{\text{line}}$ is the optimized capacity of the line or link and $p_{\text{line, max}}$ is the maximum allowable capacity of the line or link. The minimum capacity is set to the pre-existing capacity as it is assumed that the existing transmission infrastructure will not be decommissioned due to the trouble of digging up existing lines. In order to avoid transmission bottlenecks, and find the optimal solution of installed generation capacity in the nodes, the maximum capacity was set to an arbitrarily high value (1 Billion MW).

A similar approach was used for the *fossil-based* generators in the system. These generators are allowed to decommission in order to meet the decarbonization targets, but are not allowed to expand their capacity beyond the existing capacity as can be seen in equation (8) below. This was done in order to maintain a degree of realism in the model as it is unlikely that new fossil-based generators will be built in a decarbonizing energy system.

$$p_{\text{nom}} \leq p_{\text{existing}} \tag{8}$$

Most of the renewable generators are allowed to expand their capacity between 0 and an arbitrarily high value (1 Billion MW) as these are the capacities that will later be constrained by the decentralization measures. However some renewable generators like nuclear, geothermal and biomass are, constrained to "only" expand a factor 100 beyond their existing capacity int the nodes where they are already present. This is primarily done as calculating the maximum installable capacity for these technologies is a large task in itself and is outside the scope of this master's thesis. The workaround was then to only allow the expansion up to a certain limit and only in nodes that have already proven their capabilities in utilizing said technologies. The freely expandable renewable technologies in the system are therefore solar PV, onshore wind and offshore wind which are free to expand and contract with every optimization, as long as the capacity is positive as seen in equation (9) below:

$$0 \leq p_{\text{nom}} \tag{9}$$

In addition to the capacity constraints, the operation of the different generators is controlled by availability factors. Each generator's maximum power output at any at any time is given as ($p_{\text{max\_pu},g,n,t}$) and the minimum stable generation ($p_{\text{min\_pu},g,n,t}$). This can either come from technological limitations or capacity factor time series for renewables.

### *3.6. Energy storage infrastructure*

Finally, the last step before the system is ready to be optimized, is to implement the energy storage infrastructure. The pre-compiled network comes with a few different types of storage already implemented: Pumped-Hydro Storage (PHS), and Regular (reservoir) hydro storage. Historically hydro based storage in the European Union has been the primary method of energy storage, and as a result most areas capable of having hydro based storage installed have already been utilized. Therefore the hydro based storage technologies are not allowed to change their capacity in the optimization and are constrained as seen in equation (10) below. Again it is assumed that decommissioning existing hydro storage is not in the interest of the system.

$$p_{\text{nom}} = p_{\text{existing}} \tag{10}$$

However as the system decarbonizes, and the amount of dispatchable fossil-based generation decreases, it is increasingly necessary to have large amounts of energy storage capacity installed. In order to properly model this two additional storage technologies, namely battery storage and hydrogen storage, have been added to the network. Battery storage consists of batteries and inverters and hydrogen storage consists of a hydrogen tank, an electrolyzer as well as a fuel cell. When adding the additional storage technologies their costs have been calculated in a manner similar to that of the generators as seen in section 3.4 above. The cost assumptions for battery and hydrogen storage can be seen below in Table 2:

Table 2: Storage technology cost assumptions used in the optimization model [20].

| **Component** | **Type** | **CAPEX (EUR/unit)** | **Lifetime (years)** | **Efficiency** | **Annualized Cost (EUR/unit/a)** |
|---|---|---|---|---|---|
| Battery | Energy (Cells) | 142 kWh | 25 | — | 12.2k MWh/yr |
| | Power (Inverter) | 160 kW | 25 | 0.96 | 13.7k MW/yr |
| Hydrogen | Energy (Tank) | 57 kWh | 25 | — | 4.9k MWh/yr |
| | Power (Electrolyzer) | 600 kW | 30 | 0.66 | 48.4k MW/yr |
| | Power (Fuel Cell) | 1100 kW | 10 | 0.50 | 156.6k MW/yr |

Here it is important to note that the storage units include efficiencies for charging and discharging. This efficiency is incorporated in optimization model as it has a large impact on system cost, which was annualized as per equations in section 3.4 above. As for the complete cost assumptions for the different storage components these can be found in Appendix B: Storage cost assumptions.

### *3.7. Optimization strategy*

Optimizing complex energy networks, with many variables and constraints, is computationally very intensive. For this reason the optimizations performed for this master's thesis have exclusively been performed on a High Performance Computing (HPC) environment. More specifically the UCloud HPC operated by SDU [21]. The optimizations were typically run on a set of 4 machines, each with an Intel Xeon Gold 6130 CPU and took on average 12 core-hours apiece to achieve an optimal solution. The optimization algorithm was chosen as the Barrier method in the commercial optimizer by Gurobi, chosen primarily for its numerical stability [22].

### *3.8. Baseline system*

As seen in Figure 1 above, in order to be able to analyze the impacts of varying decarbonization constraints, and decentralization measures, a baseline system that is subjected to neither decarbonization nor decentralization constraints must first be optimized. This system, which will be used for comparison is referred to as the *baseline* system. The optimization of this system is performed based on the objective function and load-balance, and is a *least cost* system, meaning that the total system cost is minimized.

In order to optimize the networks first an objective function must be defined. An objective function is, as the name suggests, the function which when minimized or maximized leads to the objective of the optimization to be achieved. The objective function for finding the cheapest possible electrical system is defined as minimizing the sum of all the costs:

$$\min\Big(\sum_n \text{Capital costs} + \sum_{n,t} \text{variable costs}\Big) \tag{11}$$

Here $n$ denotes the set of all nodes, and $t$ denotes the set of all snapshots during the timeframe. The capital costs reflect things such as the cost of building generation, storage and transmission capacity and the variable costs reflect things such as fuel costs and maintenance. In order to construct a feasible solution space the objective function must be constrained by a set of bounding constraints. Initially an equality constraint is defined so that, for each snapshot, all nodes must have their electricity demand met by a combination of energy production and nodal balancing:

$$\text{generation}_{n,t} + \text{balance}_{n,t} = \text{demand}_{n,t} \quad \longleftrightarrow \quad \lambda_{n,t} \quad \forall\, n, t \tag{12}$$

Here $\lambda$ is known as the dual-variable or the Karush-Kuhn-Tucker (KKT) multiplier associated wth the constraint. This variable can be interpreted as the shadow price, or marginal price of electricity in node $n$ at time $t$. This marginal price represents the cost of supplying one additional unit of electricity to a node and as such is set by the most expensive generator needed to meet demand in that node at that time[23].

Finding the dual variable is done by formulating a Lagrangian function that relates the gradient of the objective function (11) to the gradients of the constraint (12):

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{n,t} \lambda_{n,t} g_{n,t}(x) \tag{13}$$

Here $f(x)$ is the objective function, $g_{n,t}(x)$ is the constraint function and $x$ is the set of all optimization variables.The different KKT multipliers can then be found by optimizing the Lagrangian function with respect to both the optimization variables and the KKT multipliers. Then the minima, where the gradients are zero, can be found as seen in equation (14) below:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0 \quad \text{and} \quad \nabla_\lambda \mathcal{L}(x^*, \lambda^*) = 0 \tag{14}$$

Here the asterisk denotes the optimal values of the optimization variables and the KKT multipliers respectively. Optimizing the network without a decarbonization constraint, is then used to produce a baseline system that will be used for comparison in the following sections. As a final note it is very important that the constraints used in the optimization are linear as a non-linear constraint may lead to a non-convex solution space. The result of which may be the formation of local minima that the optimizer can become trapped in, leading to sub-optimal solutions.

### *3.9. Decarbonized systems*

Once the baseline system has been optimized a set of new networks at different levels of decarbonization can be defined and optimized. In addition to the objective function (11) and the load balance (12) these systems are also constrained by a decarbonization constraint as seen in equation (15) below:

$$\text{total emissions}_C \leq (1 - C \cdot \text{baseline emissions}) \quad \longleftrightarrow \quad \mu_C \tag{15}$$

Where $C$ is the level of decarbonization as a fraction of the baseline emissions (the total emissions from the baseline system) and $\mu$ is the KKT multiplier and can be interpreted as the price of emitting an additional unit of $\mathrm{CO_2}$ in the system at decarbonization level $C$. This is additionally equal to setting the $CO_2$ price equal to the dual variable and optimizing. The baseline emissions are calculated based on the emission factors of each generator technology and their respective power output as seen in equation (16) below:

$$\text{total emissions} = \sum_t \sum_n \left( \sum_g p_{g,n,t} \cdot e_g \right) \cdot w_t \tag{16}$$

Here $p$ is the power output of generator $g$ in node $n$ at time $t$, $e$ is the emission factor of generator $g$ and $w$ is the weighting factor for snapshot $t$. The emission factors for the different technologies, are pre-loaded in PyPSA and can be seen in table 3 below:

Table 3: Emission factors by carrier technology

| **Carrier** | **Emission Factor ($tCO_2/MWh_{fuel}$)** |
|---|---|
| CCGT | 0.1870 |
| OCGT | 0.1870 |
| coal | 0.3540 |
| geothermal | 0.0260 |
| lignite | 0.3340 |
| oil | 0.2480 |

Based on previous experience with optimizing energy systems, the following levels of C were chosen for the decarbonizing systems:

$$C \in \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%, 60\%, 70\%, 80\%, 90\%, 93\%, 96\%, 99\%, 100\%\} \tag{17}$$

This was done in order to achieve a wide range of decarbonization, while increasing resolution at higher levels of decarbonization, where the system tends to change more rapidly. The outcome of the the decarbonization workflow is therefore a set of 13 different networks (excluding the baseline), each at a different level of decarbonization, defined as a fraction of the baseline emissions.

### *3.10. Decentralized systems*

As discussed in the introduction, decentralization is necessary in order to ensure a fair and secure european energy system. The first step in creating a set of decentralized systems, at each level of decarbonization, is to define a measure of decentralization. Some papers like "Modeling all alternative solutions for highly renewable energy systems" by Pedersen et al. [24] use the *Gini Coefficient* of inequality of installed renewable capacity as a measure of decentralization. Consider the following plot:

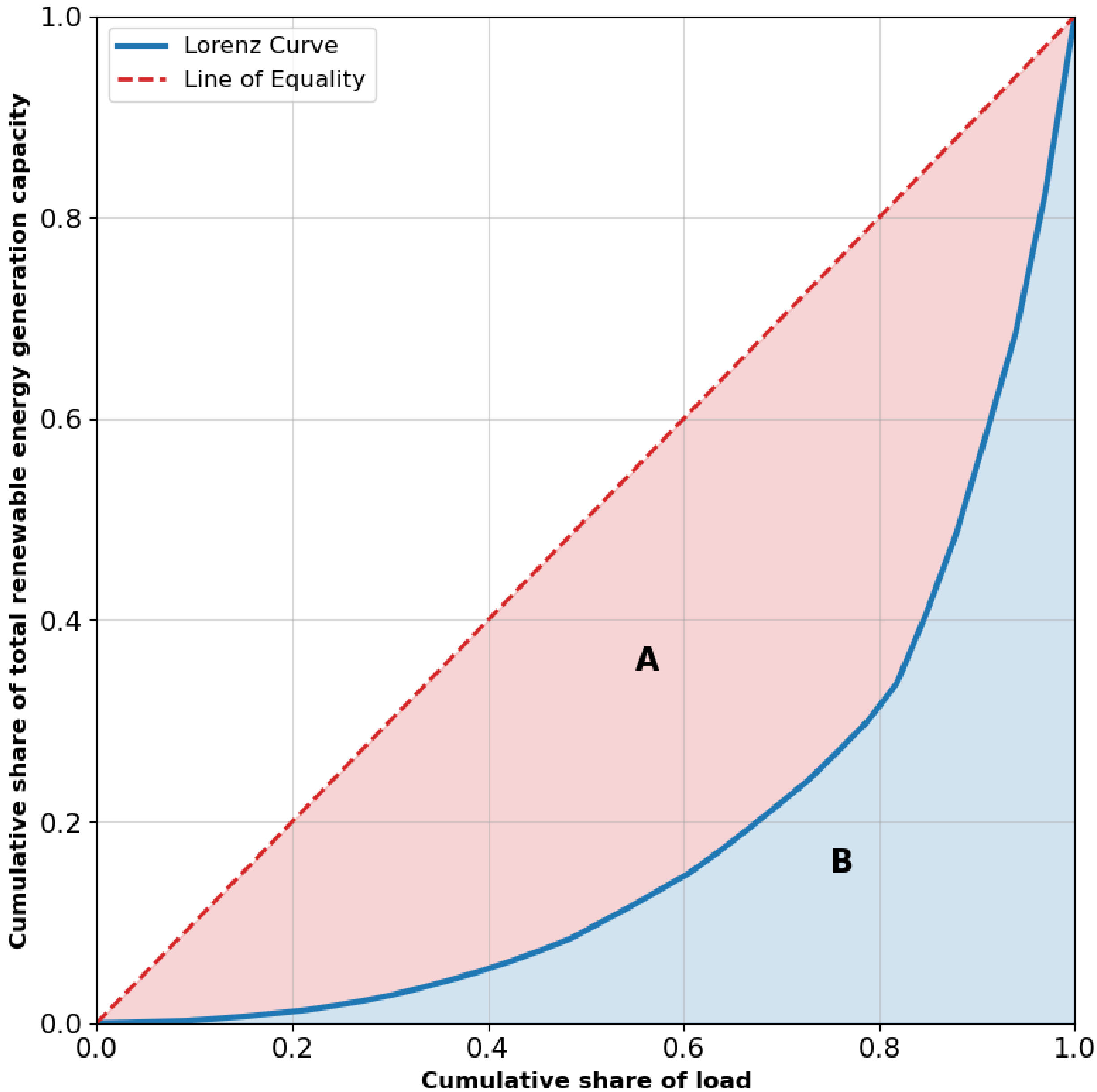


Figure 4: Example of Gini coefficient calculation for installed renewable capacity.

Here the red line represents an equal distribution of renewable capacity where the cumulative share of renewable energy generation capacity is equal to the cumulative share of the load for the same nodes. The blue line is an example of a Lorentz-curve where some nodes have a higher share of the renewable capacity than their share of the load (further to the right on the x-axis) while other nodes have a lower share of the renewable capacity than their share of the load (further to the left on the x-axis). The total inequality in the system can then be quantified by calculating the Gini coefficient as seen in equation (18) below:

$$G = \frac{A}{A + B} \tag{18}$$

Where A is the area-difference between the equality line and the Lorentz curve, and B is the area-difference between the Lorentz curve and the x-axis. This way of defining system inequality comes with a number of issues, however. Primary among these issues is that fact that the Gini coefficient is non-linear and as such cannot directly be used as a constraint as this will lead to non-convex solution spaces and local minima. While it is possible to work around this by using a sampling based approach and then calculating the Gini coefficient post optimization, this approach is difficult to understand and implement. As a result the decentralization of the systems was constrained using a novel approach based on [5] as seen in equation (19) below:

$$\frac{1}{K} \cdot \omega_n \cdot R_{\text{total}} \leq R_{\text{cap},n} \leq K \cdot \omega_n \cdot R_{\text{total}} \quad \forall n \tag{19}$$

Where $\omega_n = \frac{L_n}{\sum_{n'} L_{n'}}$ is the normalized load-share weight for node $n$, $R_{\text{total}}$ is the total renewable capacity across all nodes, $R_{\text{cap},n}$ is the installed renewable capacity at node $n$, and $K$ is the decentralization parameter. This way each node is given a minimum renewable capacity as a fraction of the total renewable capacity in the system based on their share of the total load. Simultaneously each node is also assigned a maximum renewable capacity as a factor of the total renewable capacity in the system based on their share of the total load. If, for example $K = 3$ that means that each node is allowed to "outperform" their load-share weighted renewable capacity by a factor of 3, while also being required to install at least one third of their load-share weighted renewable capacity. By then gradually having $K$ approach 1 the system is forced to decentralized until at $K = 1$ each node is required to install renewable capacity exactly equal to their load-share weighted renewable capacity, which leads to the system becoming increasingly decentralized.

Based on preliminary experiments following levels of K were chosen for the decentralized systems:

$$K \in \{7, 6, 5, 4, 3, 2, 1\} \tag{20}$$

The decarbonized system with no decentralization constraint enforced is referred to as $K_{inf}$. This approach has the benefit of being linear, but also avoids another pitfall when implementing decentralization in systems that also are subjected to decarbonization constraints. Both constraints, in some way, interact with the amount of renewable capacity installed in the nodes of the network. If one was to simply constrain the amount of installed renewable capacity in each node without taking into account the total amount of renewable capacity in the system, it could lead to the system installing more renewable capacity than needed to meet the decarbonization constraint. This interaction would then lead to the system becoming overly decarbonized and difficult to compare to other systems. It is for this reason that the decentralization constraint is defined as a fraction of the total renewable capacity in the system, ensuring that the decentralization constraint scales with the amount of renewable capacity needed to meet the decarbonization constraint. As a side note the upper bound on the nodal renewable capacity relative to the nodal load is defined as a fraction of the system wide-capacity to load ratio. This means that the upper bound, for installed nodal renewable capacity, is not the same as K but rather K multiplied by the system-wide load to generation capacity ratio.

The final outcome of the Snakemake workflow is therefore a set of 105 different networks, each at a different level of decarbonization and decentralization. Before proceeding to the results of the optimization workflow, it is useful to introduce a more intuitive measure of the decentralization of a given system as the K parameter is somewhat abstract. A more intuitive measure of decentralization is the *Standard Deviation of Nodal Renewable Capacity per Load* defined as:

$$\text{SDRCL} = \sqrt{\frac{1}{n}\sum_{n}\left(\frac{R_{\text{cap},n}}{L_n} - \mu\right)^2} \quad \text{where} \quad \mu = \frac{1}{n}\sum_{n}\frac{R_{\text{cap},n}}{L_n} \tag{21}$$

Here $R_{\text{cap},n}$ is the installed renewable capacity at node $n$, $L_n$ is the load at node $n$ and $n$ is the number of nodes. This measure represents the standard deviation of the ratio between installed renewable capacity and load at each node. A low value of SDRCL indicates that the renewable capacity is more evenly distributed across the nodes relative to their load, while a high value indicates a more uneven distribution. By then comparing the SDRCL values for each of the decentralized systems to the network at the same level of decarbonization but without decentralization constraints, it is possible to get a more intuitive measure of how decentralized a given system is and how successful the decentralization constraints were.

## 4. Results

In the following section the results from the workflow discussed in the methodology section will be shown and analyzed. Starting with the baseline system, followed by the decarbonized systems and finally the decentralized systems. After each type of system the underlying mechanisms and interactions present in the systems will be discussed.

### *4.1. Baseline system*

As a reminder the baseline system is purely constrained by the objective function (11) and the load balance constraint (12). Figure 5 below shows a map of the installed generation capacities for the baseline system:

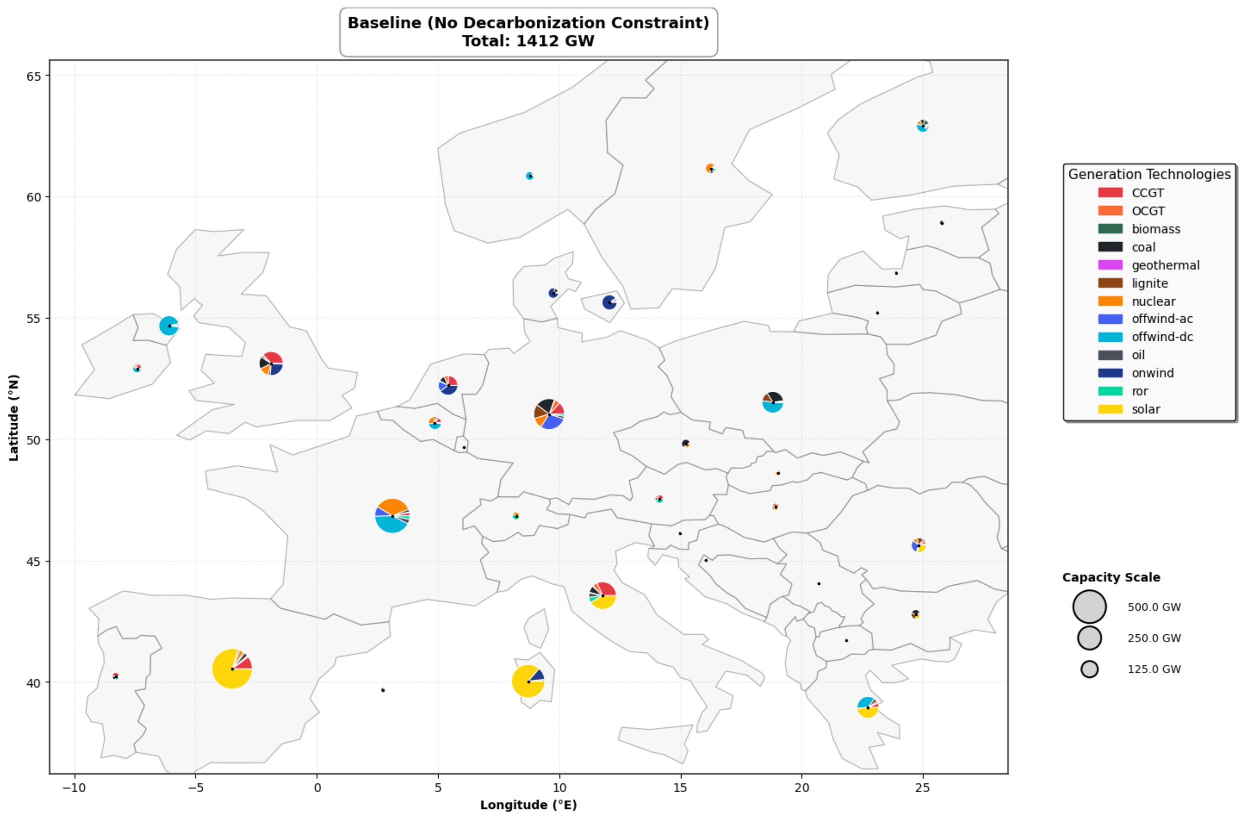


Figure 5: Baseline generator capacity map.

Here it can be seen that the mismatch between the load and the fossil generation has been filled with renewable generation, as the fossil based generation technologies are not allowed to expand their capacity. The total installed generation capacities in the baseline system, as well as the annual generation can be seen in figure 6 below:

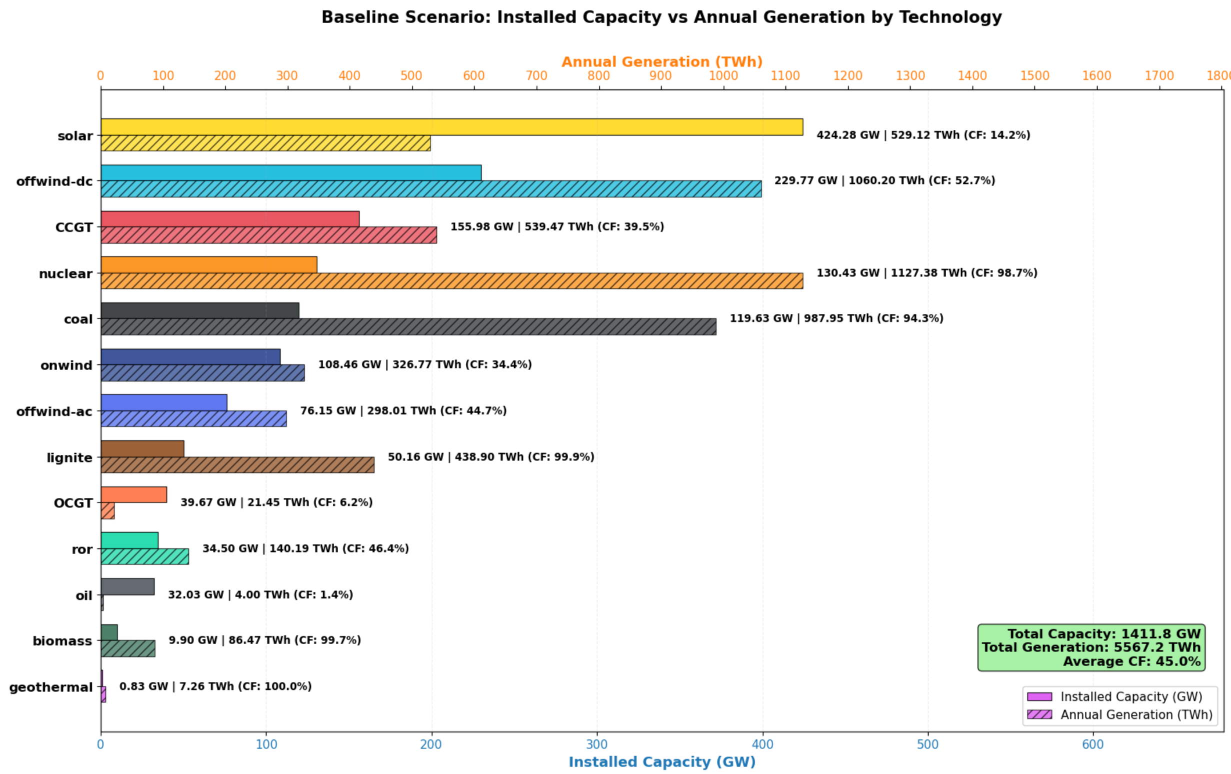


Figure 6: Baseline generator capacity and annual generation by technology.

As seen in figure 6 above the installed generation capacity is dominated by solar PV, which makes up almost a third of the total generation capacity. However as technologies differ in capacity factor time series and utilization, the actual generation by technology follows a different pattern.

The generation mix still has a large fraction provided by fossil based technologies and the largest contributor is nuclear power. This is caused by its dispatchable nature, which does not require storage to the same extent as weather dependent generation. Renewable generation ends up providing a bit over half of the total electricity. As for the transmission capacities these can be seen for the baseline scenario in figure 7 below:

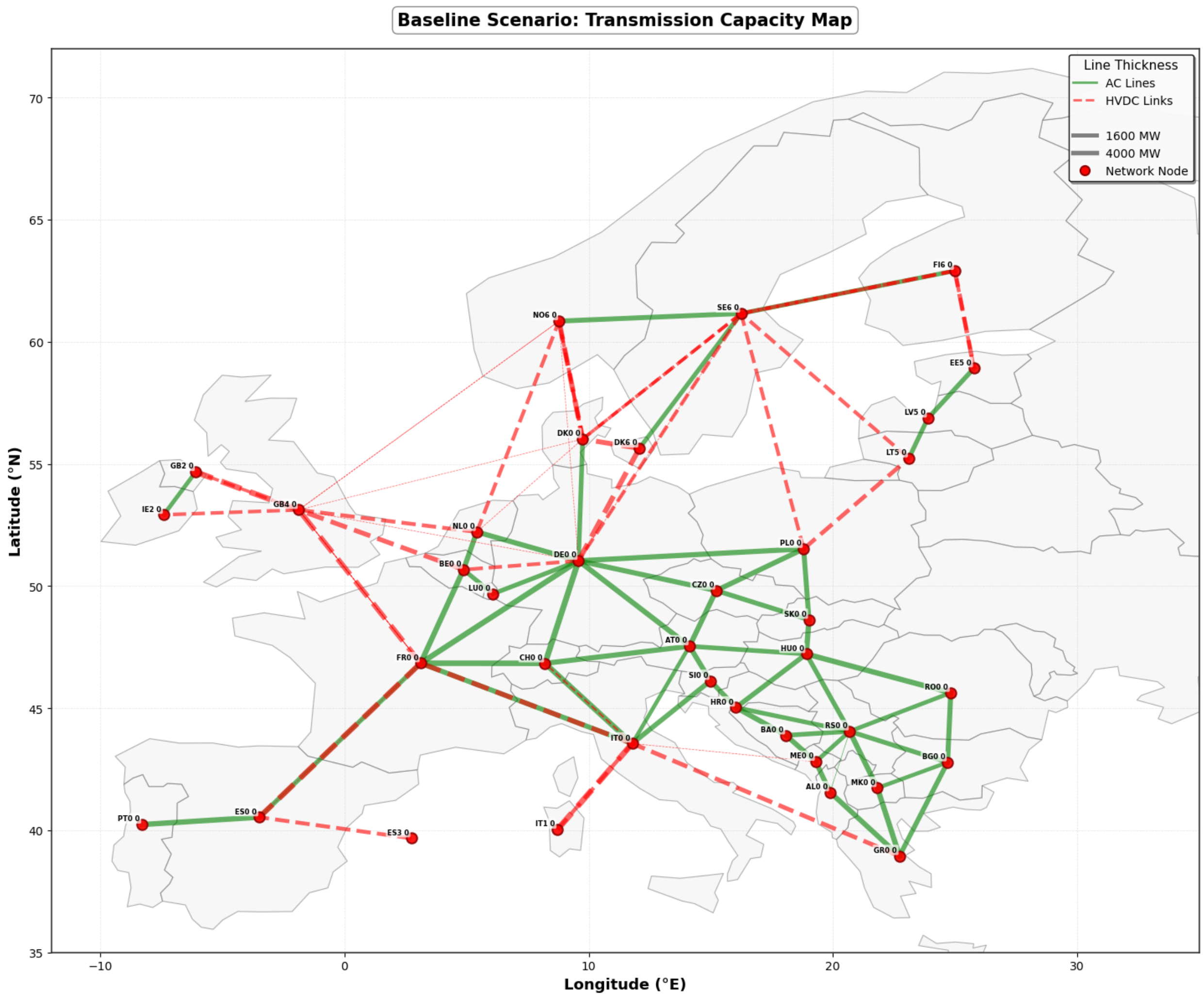


Figure 7: Baseline transmission capacity map.

The transmission infrastructure in the baseline system contains 490.2 $GW$ HVDC capacity and 232.4 $GW$ AC capacity. Since a large amount of the load is now serviced by renewable generation, whose capacity factor time series do not necessarily line up with the load time series, storage infrastructure and transmission is used to de-couple supply and demand temporally. The installed storage capacities totalling 650 $GW$ in the baseline system can be seen below in figure 8:

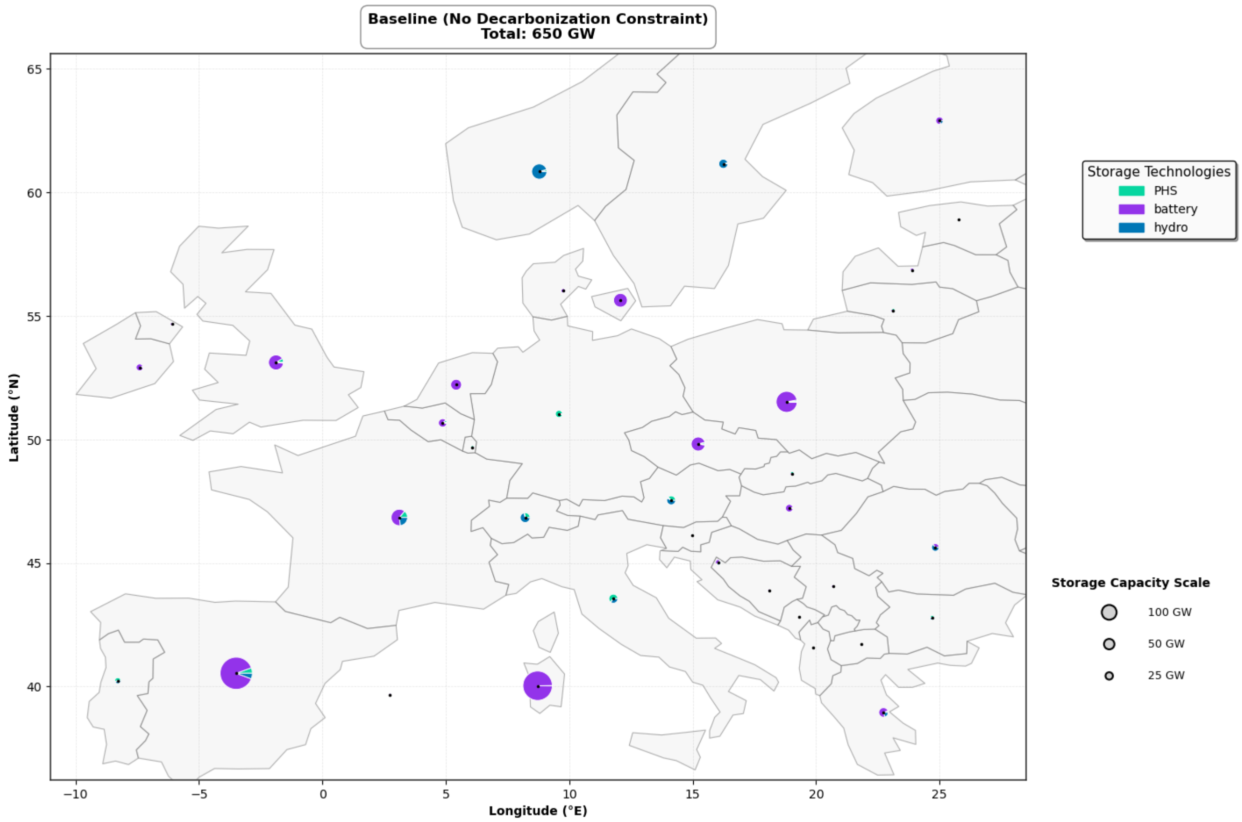


Figure 8: Baseline storage capacity map.

Here it can be seen that the optimization resulted in a very substantial buildout of battery storage capacity, while the hydrogen storage capacity is absent. This is primarily caused by the high round trip efficiency loss as seen in table 2 above. This large loss leads to hydrogen based storage becoming prohibitively expensive compared to battery storage. The lack of hydrogen based storage also means that the system is less likely to utilize wind-based renewable generation compared to solar based generation as wind generation tends to be more seasonal, and therefore better served by hydrogen storage.

The total emissions from the baseline system as calculated by equation (16) is 1297 $MtCO_2$. The breakdown of which technologies contribute to these emissions can be seen below in figure 9:

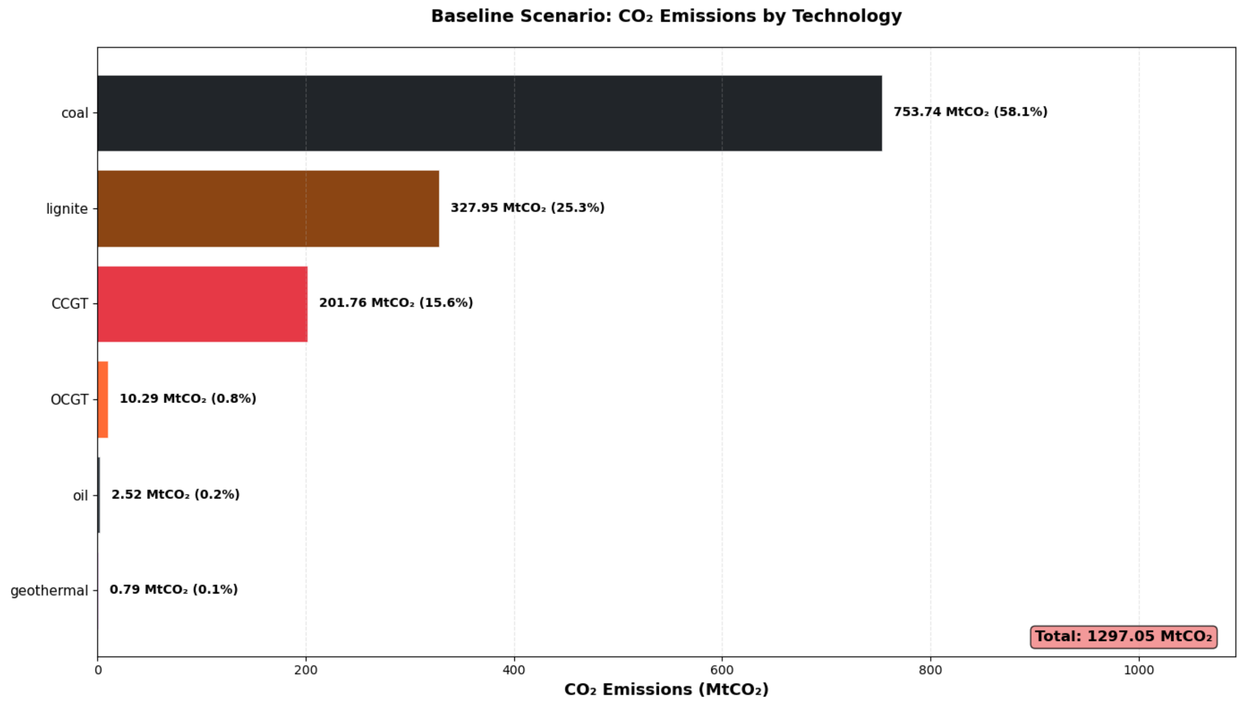


Figure 9: Baseline emissions by technology.

Here the two solid-fuels coal and lignite are the major contributors to the emissions. This is to be expected when considering their high emission factors as seen in table 3 and their large generation contributions as seen in figure 6 above.

For the economic side of the baseline system the mean marginal demand-weighted price and the total system cost are of interest. These metrics can be seen below in table 4:

Table 4: Baseline System Economic Metrics (System Cost vs Market Price)

| **Metric** | **Value** |
|---|---|
| Mean Marginal Price (Demand-Weighted) | 68.56 €/MWh |
| Total System Cost | 208.99 B€/year |

Here The mean marginal demand-weighted price is defined as seen in equation (22) below:

$$\text{Mean Marginal Demand-Weighted Price} = \frac{\sum_{t \in T} \sum_{n} (L_{n,t} \times \lambda_t)}{\sum_{t \in T} \sum_{n} L_{n,t}} \tag{22}$$

For time $t$, Nodes $n$, Loads $L$ and marginal price $\lambda$. This cost measure represents the average price that consumers pay for electricity over the entire time period, weighted by their demand. The total system cost is simply the sum of all costs as minimized by the objective function (11). The total system cost represents the planner's perspective on the cost of operating the system, while the mean marginal demand-weighted price represents the consumer's perspective on the cost of electricity.

Finally, the system is, visually, as seen in figure 5 above quite centralized. This can be quantified by calculating the Standard Deviation of Nodal Renewable Capacity per Load (SDRCL) as defined in equation (21). For the baseline system the SDRCL is calculated to be 11.84 $\mathrm{MW_{renewable}}/\mathrm{MW_{load}}$. This figure is difficult to quantify in isolation as its intended use is as a comparison metric. However by visualizing the nodal renewable capacity per load in figure 10 below it is possible to get an idea of how for centralized the baseline system is.

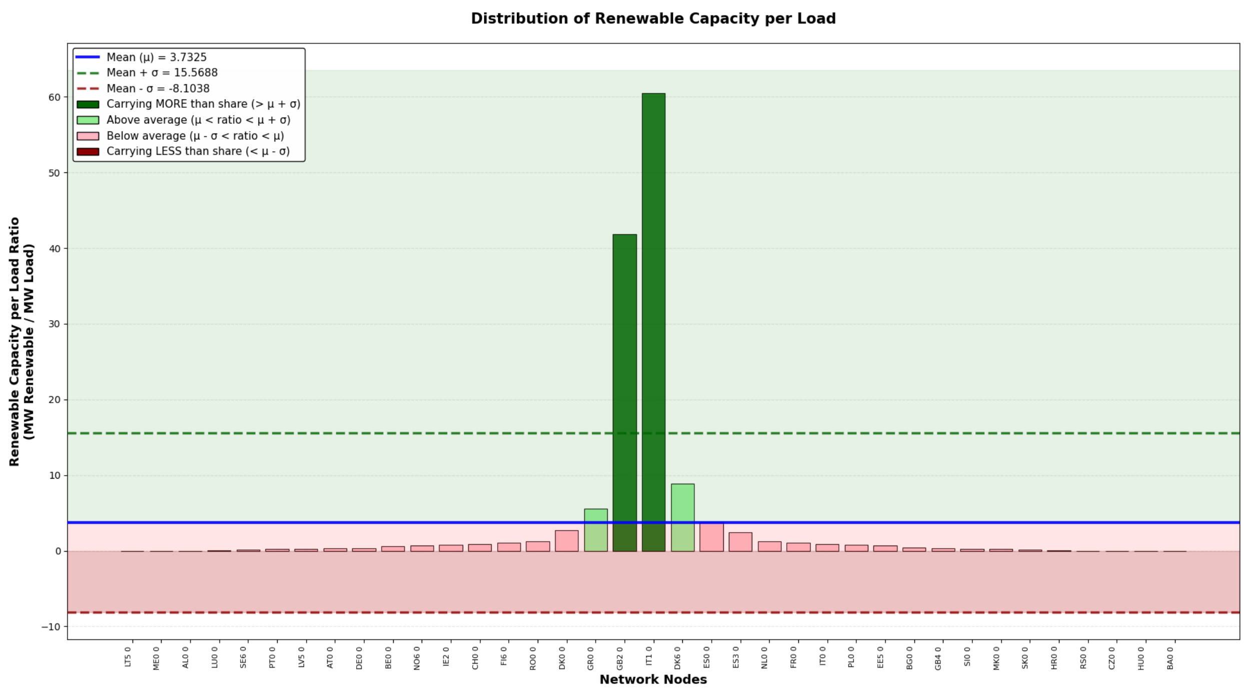


Figure 10: Baseline renewable capacity per load map.

Here it can be seen that only four nodes (IT0, GB2, GR0 and DK6) have a larger installed renewable capacity than their yearly load share. The relevant statistics for the centralization of the baseline system can be seen in table 5 below:

Table 5: Baseline System: Renewable Capacity per Load Distribution Statistics

| **Metric** | **Value** |
|---|---|
| Total Nodes (with load) | 37 |
| Mean Ratio ( $\mathrm{MW}_{\mathrm{renewable}}/\mathrm{MW}_{\mathrm{load}}$) | 3.73 |
| Standard Deviation (SDRCL, $\sigma$) | 11.84 |
| Upper Bound ($\mu + \sigma$) | 15.57 |
| Lower Bound ($\mu - \sigma$) | -8.10 |
| *Distribution:* | |
| Nodes Above $\mu + \sigma$ | 2 |
| Nodes Within $\pm\sigma$ | 35 |
| Nodes Below $\mu - \sigma$ | 0 |

### *4.2. Baseline mechanisms and interactions*

The baseline system behavior is, as can be seen above, fairly intuitive. As the pre-compiled network is a *Brownfield* system that contains existing fossil-based generation capacity, but no renewable based generation capacity, the optimizer services the load with the renewables, as the fossil based generators are not allowed to expand their capacity. The large capacity buildout is dominated by solar PV in the south of Europe and off-shore and on-shore wind in the northern regions. As for the actual generation, as can be seen in figure 6 above, the massive solar buildout has not converted into a similarly large generation contribution. This is primarily caused by the fact that solar PV is more limited by its capacity factor time series than wind based generation. The largest contributor to the generation ends up becoming nuclear power, which has a high capacity factor time series and is not limited by weather conditions. In addition nuclear, is dispatchable and therefore does not require storage to the same extent as weather dependent generation.

Transmission infrastructure has also been expanded in order to transport the energy from non-dispatchable renewables to the load centers where it is needed.

The baseline emissions, are as explained above, dominated by solid fuel based fossil generation technologies due to their high emission factors. The mean marginal demand-weighted price of electricity in the baseline system is calculated to be 68.56

€/MWh which is fairly reasonable and well within the same order of magnitude as found in papers on similar models such as in [5].

Finally the baseline system as seen both visually in figure 10 perfectly encapsulates the need for decentralization measures. While the heavily centralized nature of the system is optimal from a simple least-cost perspective, it will likely to lead to issues with public acceptance, political feasibility and security.

### 4.3. Decarbonized systems

Since the outcome of the decarbonization workflow is the baseline plus 13 systems, each for a different level of decarbonization, it is more useful to visualize the evolution for the key metrics, rather than go through them one by one. Seen below is the mix of installed generation capacity for the different levels of decarbonization in figure 11:

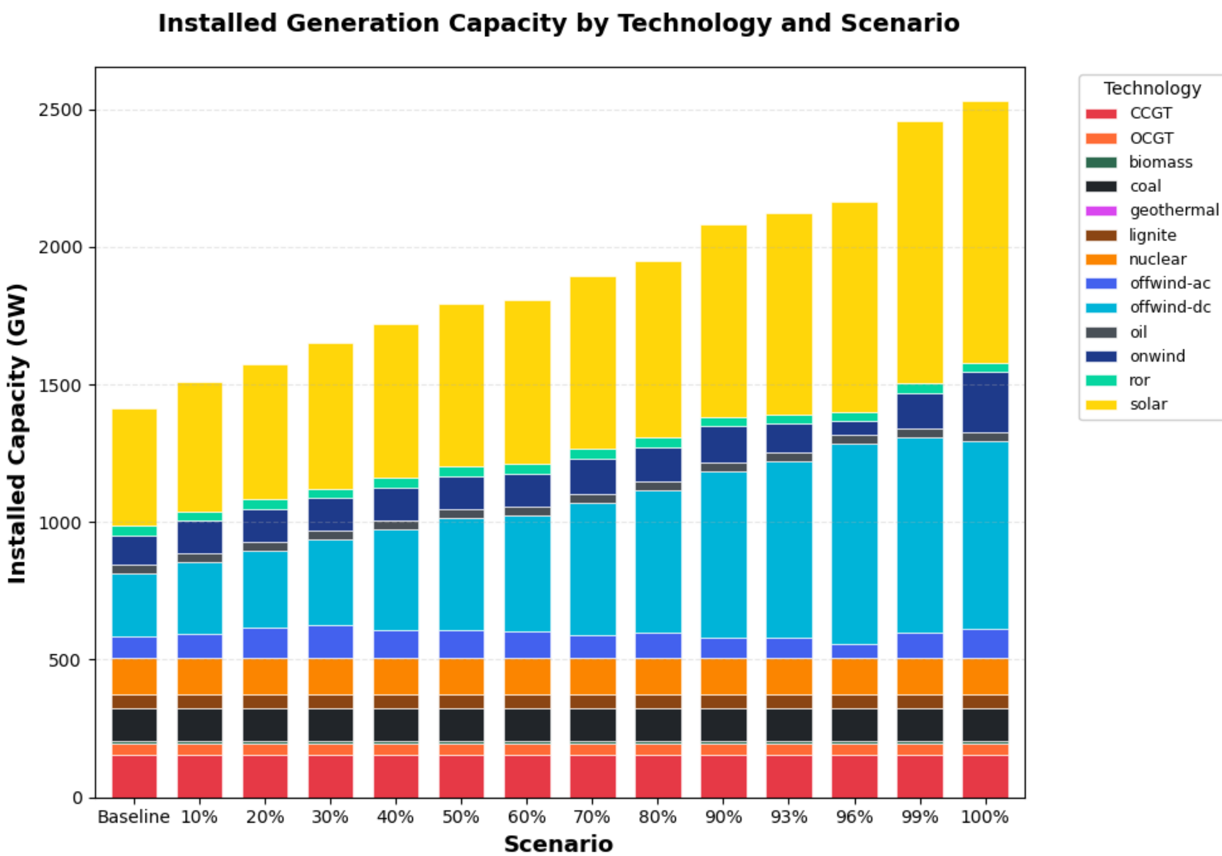


Figure 11: Decarbonized generator capacity by technology.

Here it can be seen that in order to decarbonize the system completely the total installed generation capacity increases from around 1400 $GW$ at the baseline, to around 2500 $GW$ in the fully decarbonized system, or a around a 78% increase. As expected the fossil based generation capacity does not disappear as the system is not forced to decommission existing capacity, but rather to replace the actual generation from fossil based generators with renewable generation. The actual generation breakdown for the different levels of decarbonization can be seen below in figure 12:

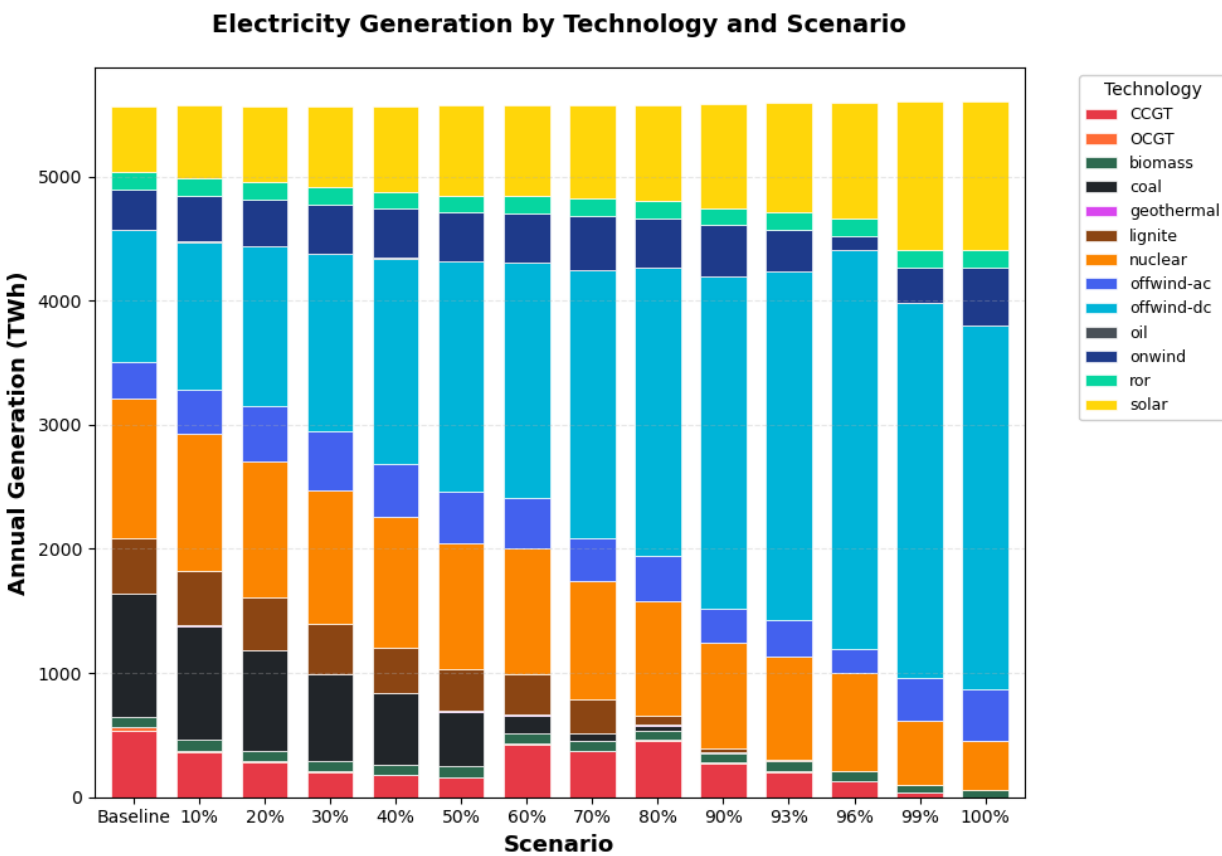


Figure 12: Decarbonized generator output by technology.

As seen in figure 12 above when the system decarbonizes the fossil fuel based electricity generation goes to zero and is replaced primarily with offshore wind and solar PV generation. In order to better visualize the evolution of the generation of the different technologies as the system decarbonizes, see figure 13 below.

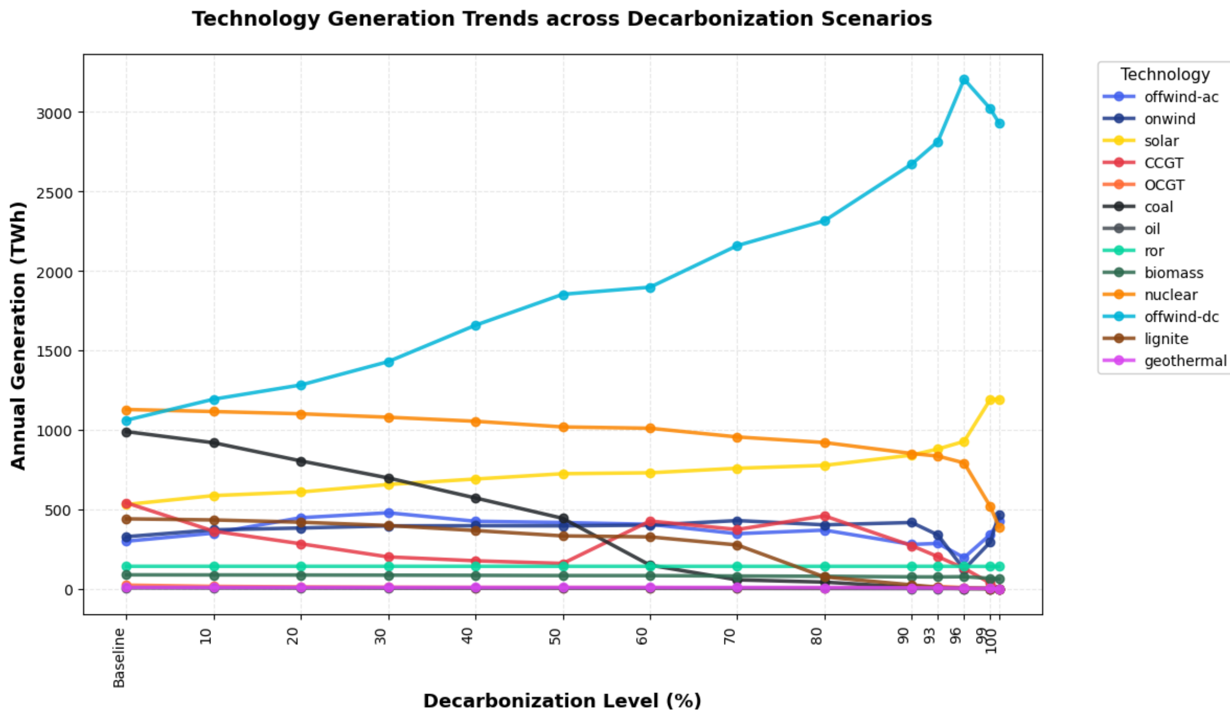


Figure 13: Decarbonized generator output by technology (line plot).

Here the generation dominance of off-shore wind at high levels of decarbonization is clearly visible. Interestingly at very high levels of decarbonization nuclear generation, as well as wind and biomass declines. In figure 13 above there seems to be a trade-off at around 96% decarbonization where the system suddenly priorities DC-offshore wind instead of AC-offshore generation. Then at even higher levels of decarbonization this tradeoff then gradually disappears, and the levels at 100% decarbonization end up being fairly consistent with the rest of the trend-line. This seems to be an artifact of the discrete steps taken in the modelling of the decarbonization results. As the $CO_2$ constraint tightens the viable solution space shrinks which can lead to sudden shifts in the optimal solution as seen here. On the other hand the sudden decline in the amount of nuclear generation at high levels of decarbonization is not an artifact, but an interesting mechanism. The sudden decline in nuclear generation coincides with the large buildout of the storage capacity and seems to indicate that the system transitions from being dominated by one paradigm to being dominated by another paradigm. Nuclear generation is dispatchable and such has primarily acted as a way of balancing the system when the non-dispatchable renewables are not producing enough electricity. As the system is forced to build storage by some other mechanism at high levels of decarbonization less nuclear generation is needed for balancing, which leads to the decline. In regards to the the transmission infrastructure, in the decarbonized systems, the total transmission capacity seen below in figure 14:

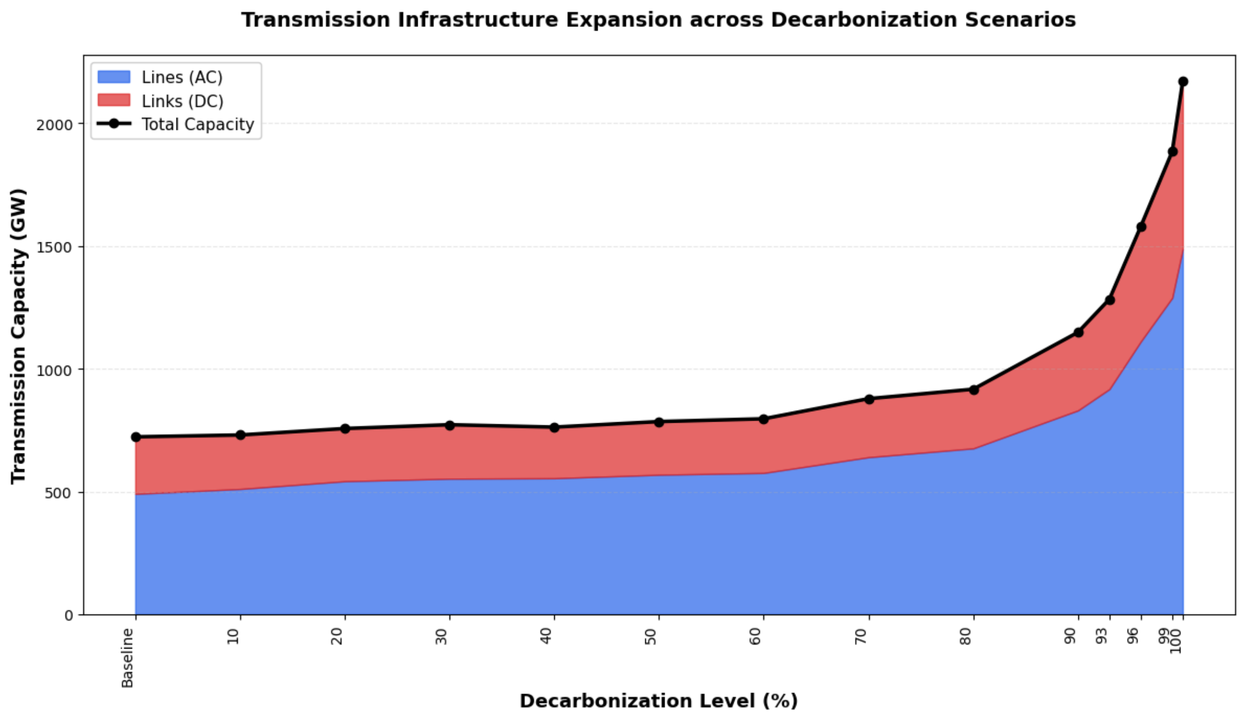


Figure 14: Decarbonized transmission capacity by technology.

The contribution to the total transmission capacity from the HVDC links remains fairly constant as the system decarbonizes, while the AC transmission capacity increases substantially. This is due to the fact that HVDC links are mainly used for long-distance or cross-ocean transmission, while the AC lines are mainly used in mainland Europe as a part of the main continental mesh Network. As the system decarbonizes the optimizer prioritizes the expansion of the AC part of the network, as this is what benefits more nodes in the system. This then in turn provides more flexibility in the location of storage and generation compared to expanding the HVDC network. However at the very extreme of 100% decarbonization the HVDC capacity does see a small increase as the system now needs to use the renewables in the best resource locations regardless if this in mainland Europe or, for example, in the UK where offshore wind resources are very good. As expected the total transmission capacity sees a large increase at very high levels of decarbonization as the system relies more and more on non-dispatchable renewable generation which must be transported from the generation sites to the load centers or stored. The storage capacity evolution for the decarbonized systems can be seen below in figure 15:

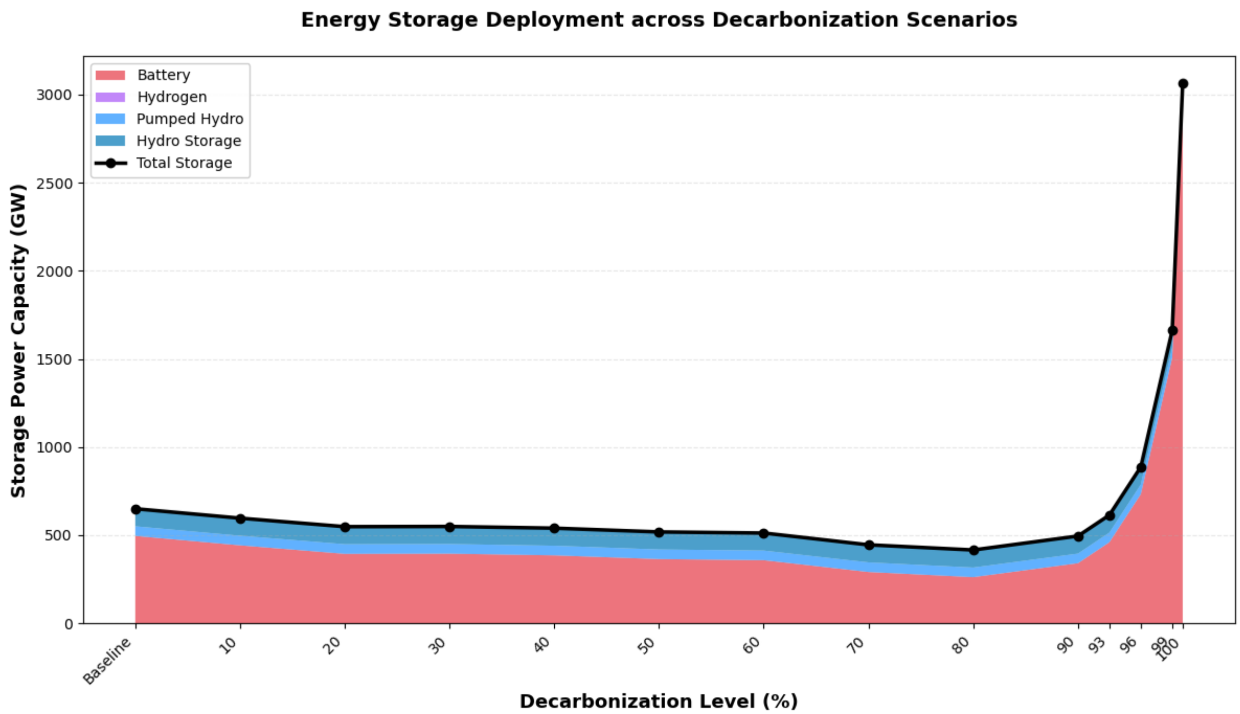


Figure 15: Decarbonized storage capacity by technology.

As alluded to above a massive increase in storage capacity is seen at at the very highest levels of decarbonization. This is caused by the decommissioning of the last dispatchable generation technologies which forces the system to use storage for nodal balancing where it has previously been able to rely on dispatchable fossil fuel based generation. Again the complete absence of hydrogen storage is notable, especially as the generation share of wind based generation is very high. This is not very intuitive as one typically associates hydrogen storage with long-term storage and wind based generation which is more seasonal in nature. However it seems that the poor round-trip efficiency of the hydrogen storage means that it is not cost competitive with battery storage even at high levels of wind based generation. The rate of which wind based generation is built out trends towards wind-based generation being prioritized increasingly at higher levels of decarbonization. As will be explained later in this section, this also coincides with the sharp increase in scarcity rent of hydro storage as seen in figure 18 below. This can be interpreted as the fact that, if unconstrained, the optimizer would have increase water-based storage capacity which is a form of long-term storage and as such is better suited to balance wind based generation.

The emissions of the decarbonizing systems are of course fractions of the baseline emissions as constrained by equation (15). The mean marginal demand-weighted price as defined by equation (22) can be seen in figure 16 below and the total system cost evolution for the decarbonized systems can be seen below in figure 20:

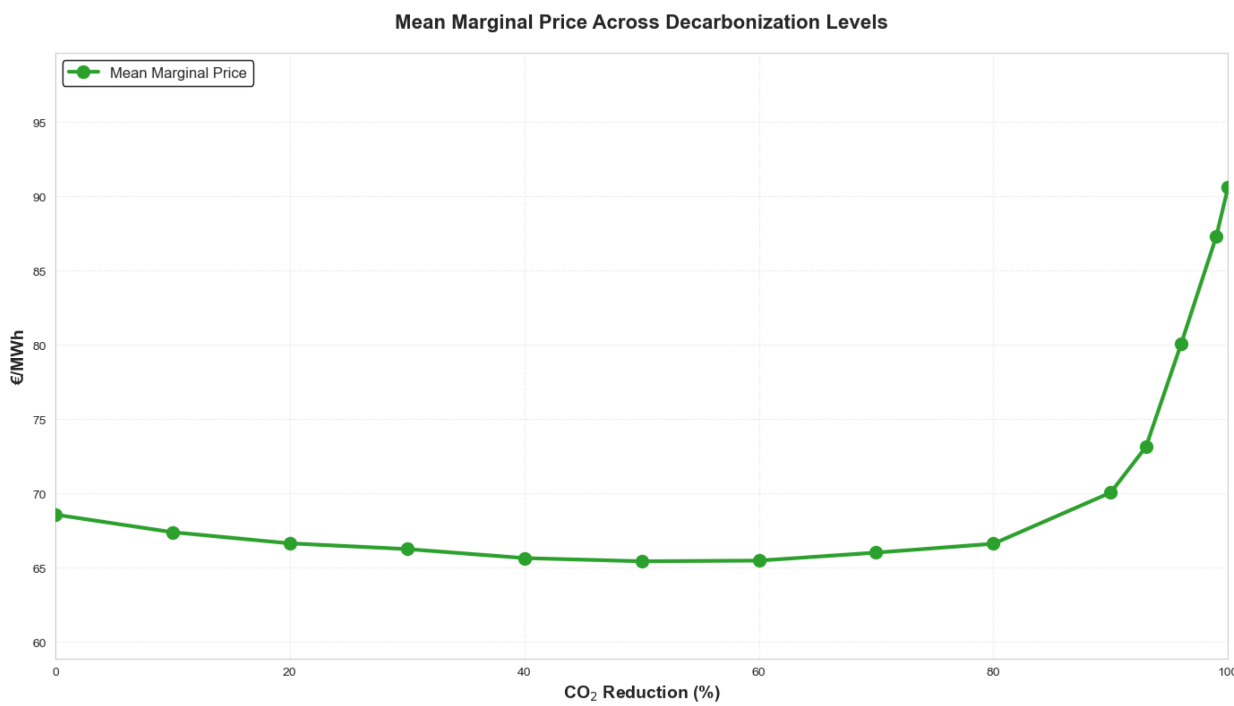


Figure 16: Decarbonized system economic metrics.

Interestingly the mean marginal demand-weighted price initially *decreases* as the system begins to decarbonize before increasing dramatically at high levels of decarbonization. While counterintuitive in a least cost optimized system, this is in fact not an artifact of the model but rather a result of the underlying mechanisms at play in the system. When reviewing the electricity prices associated with the optimized networks, it is of interest to compare two different cost measures: The mean marginal demand-weighted price and the levelized cost of electricity (LCOE). The levelized cost of electricity (LCOE) is defined as:

$$\text{LCOE} = \frac{\sum_{t \in T} \sum_{g} \left(\text{Capital Costs}_{g,t} + \text{Marginal Costs}_{g,t}\right)}{\sum_{t \in T} \sum_{n} L_{n,t}} \tag{23}$$

For generator technologies $g$. LCOE represents the average cost of producing electricity over the lifetime of the system.

Seen below in Figure 17 is a plot comparing the mean levelized cost of electricity for the different levels of decarbonization, compared to the mean marginal demand-weighted price:

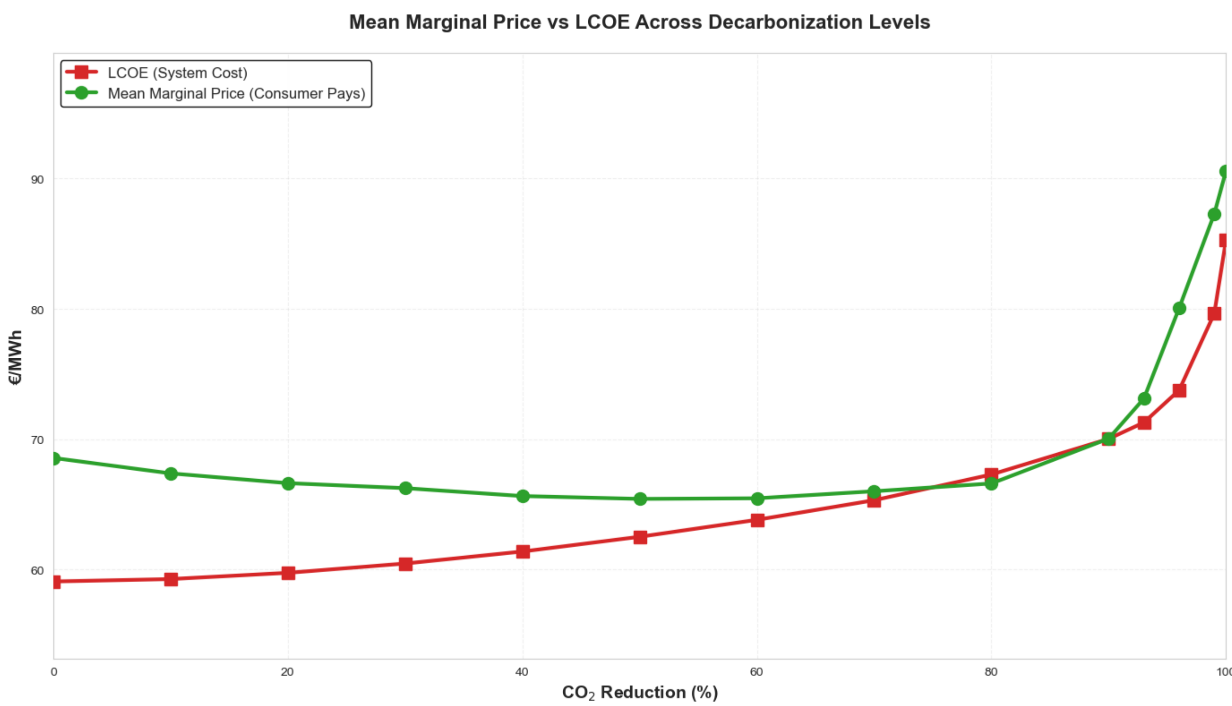


Figure 17: Comparison of mean LCOE and mean marginal price

This plot shows two different interesting results. Firstly, as commented on previously, it can be seen that as the system begins to decarbonize, initially the mean marginal demand weighted price *decreases* before increasing dramatically at high levels of decarbonization. Secondly it can be seen that for a majority of decarbonization levels a significant gap between the two cost measures is present.

Traditionally, optimized energy system models such as the ones produced by PyPSA, must adhere to the zero-profit condition [25]:

$$\sum_{t \in T} \sum_{n} (L_{n,t} \times \lambda_t) = \sum_{t \in T} \sum_{g} \big(\text{Capital Costs}_{g,t} + \text{Marginal Costs}_{g,t}\big) \tag{24}$$

This equation states that for all times the demand-weighted marginal price in all nodes must be equal to the sum of the capital and marginal costs across all technologies and times. Or, in other words: The revenue must be equal to the expenditure. However, critically, the zero-profit condition applies to long-run competitive equilibrium subject to the following assumptions:

- All capacity is free to expand and contract in order to achieve optimality.
- No artificially imposed constraints on the system.
- Steady state market conditions.

This clashes with the modelling methodology of this thesis in several ways. The optimized model is a *Brownfield* system, meaning that even before optimization some fossil generation capacity exists in the system. As stated in the modelling section 3.5 , fossil generation capacity is allowed to contract but not to expand as constrained by:

$$p_{\text{nom}} \leq p_{\text{nom, max}} = p_{\text{nom, baseline}} \tag{25}$$

This means that the zero profit condition does not apply to optimization performed in this thesis, which leads to the difference between levelized cost and mean marginal price, also known as scarcity rent, as seen in Figure 17. The scarcity rents manifest when the optimizer ideally would increase capacity to meet demand, but is constrained by a binding capacity constraint.

Imagine for example that a fossil fuel-based generator is producing electricity at a lower marginal cost than the current market price. If the fossil based generator was allowed to expand, this would naturally attract additional fossil based generation to the market, until it becomes saturated and the mean marginal demand weighted price decreases to the point, where the fossil generator is no longer profitable. However, as the fossil generator is not allowed to expand capacity, it can continue to produce electricity at a profit, leading to the scarcity rents seen in the system.

This is the primary driving mechanism behind the gap between LCOE and mean marginal demand weighted prices seen in Figure 17.

This mechanism also applies to non-extendable renewable generation capacity, like nuclear, geothermal and biomass, as well as non-extendable storage like pumped hydro-storage. Pictured below, in Figure 18 is the evolution of the contribution to the total scarcity rent from the different technologies not adhering to the zero-profit condition:

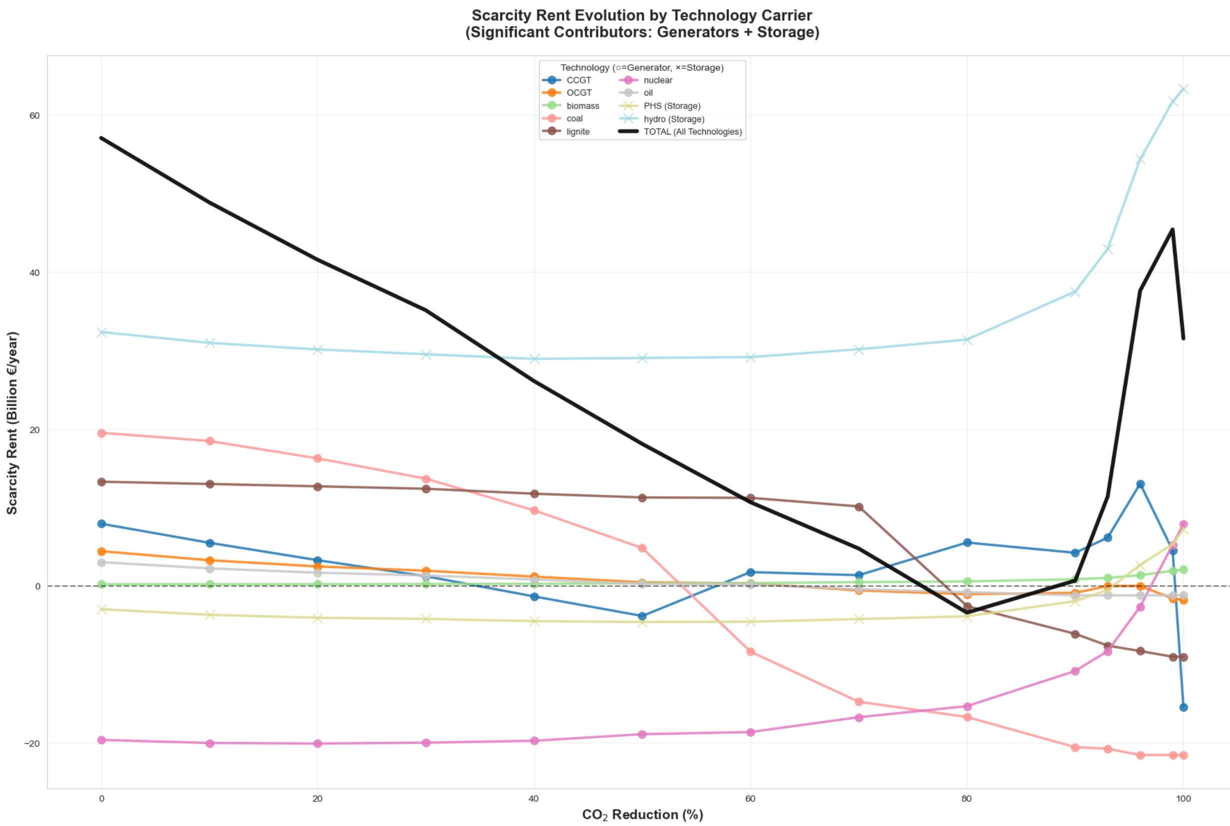


Figure 18: Scarcity rent contributions by technology

Here it can be seen that at low levels of decarbonization the majority of the contribution to the total scarcity rent comes from the fossil-fuel based generators. As the system decarbonizes the amount of fossil fuel generation is reduced, in order to meet the decarbonization constraint, and as such the contribution to the scarcity rent decreases. However at around 80% decarbonization, where the vast majority of the fossil based generation has been phased out, the main contributors become the dispatchable renewable generators, such as nuclear, and storage units needed for balancing. This is illustrated in the modified version of Figure 17 (Figure 19) below:

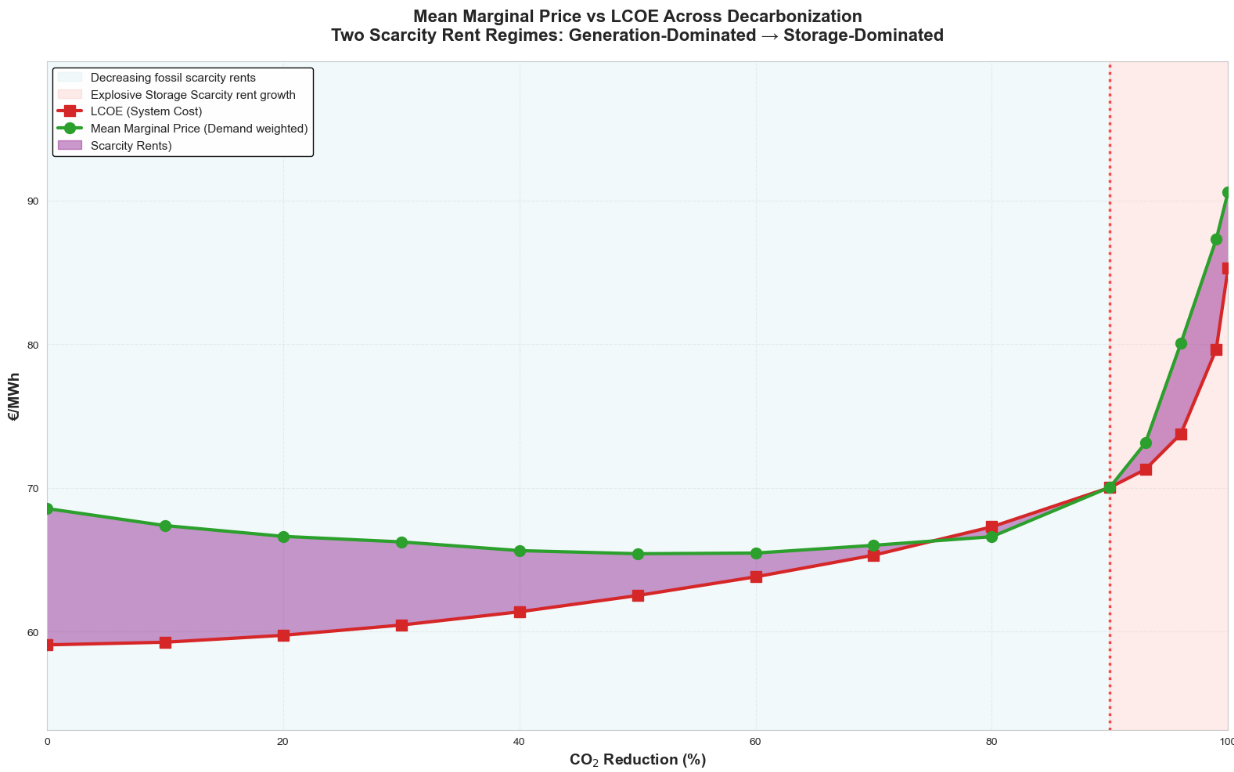


Figure 19: Transition between main scarcity rent contributors

As a side note, a negative scarcity rent indicates that the technology in question is running at a loss, meaning that the revenue from selling electricity is not sufficient to cover the capital and marginal costs of the technology. One might think that such a technology would not be included in the optimized system, however as the optimization is performed on a brownfield system, with existing capacity, it may be optimal to keep such a technology running at a loss if the overall system cost is reduced by doing so.

As for the initial decrease in mean marginal demand weighted price. This can be attributed to something called the "merit-order effect": When the decarbonization constraint of fossil fuel energy generation tightens, fossil-fuel based energy production is replaced by renewable generation. As some forms of renewable generation have very low marginal costs the mean marginal demand weighted price decreases, as these low marginal cost generators set the market price in more snapshots. However the total system cost increases as the newly installed generation- and storage-capacity has capital costs that contribute to the overall increase in system cost as seen below in figure 20:

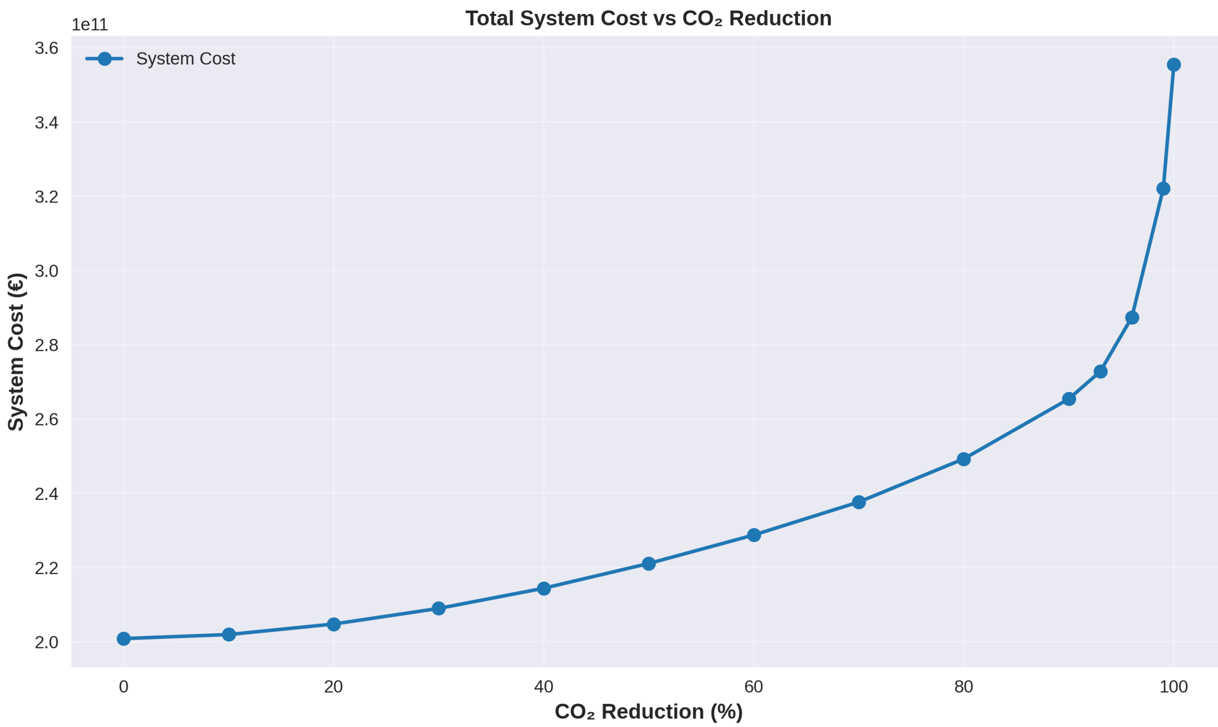


Figure 20: Total system cost in least-cost systems

The total system cost increases quite dramatically from 200 B€ at the baseline to around 360 B€ at 100% decarbonization, corresponding to a 80% increase. The main contributor to this large increase is the large buildout of transmission- and storage-infrastructure needed for balancing as can be seen in figure 21 below.

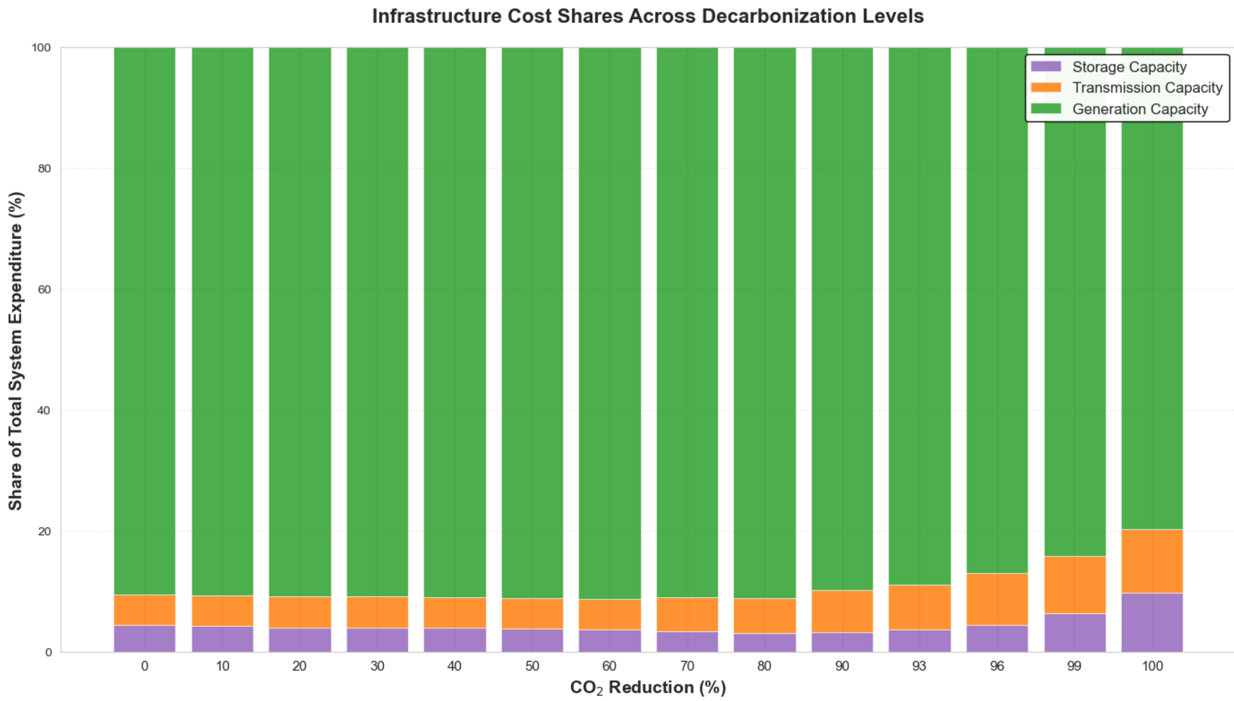


Figure 21: Decarbonized system cost share breakdown by category.

Finally the decentralization of the decarbonized systems can be quantified by the change in the equity measure (the SDRCL as defined in equation (21)) compared to that of the baseline. Seen below in figure 22 is the change of the equity for the decarbonized systems:

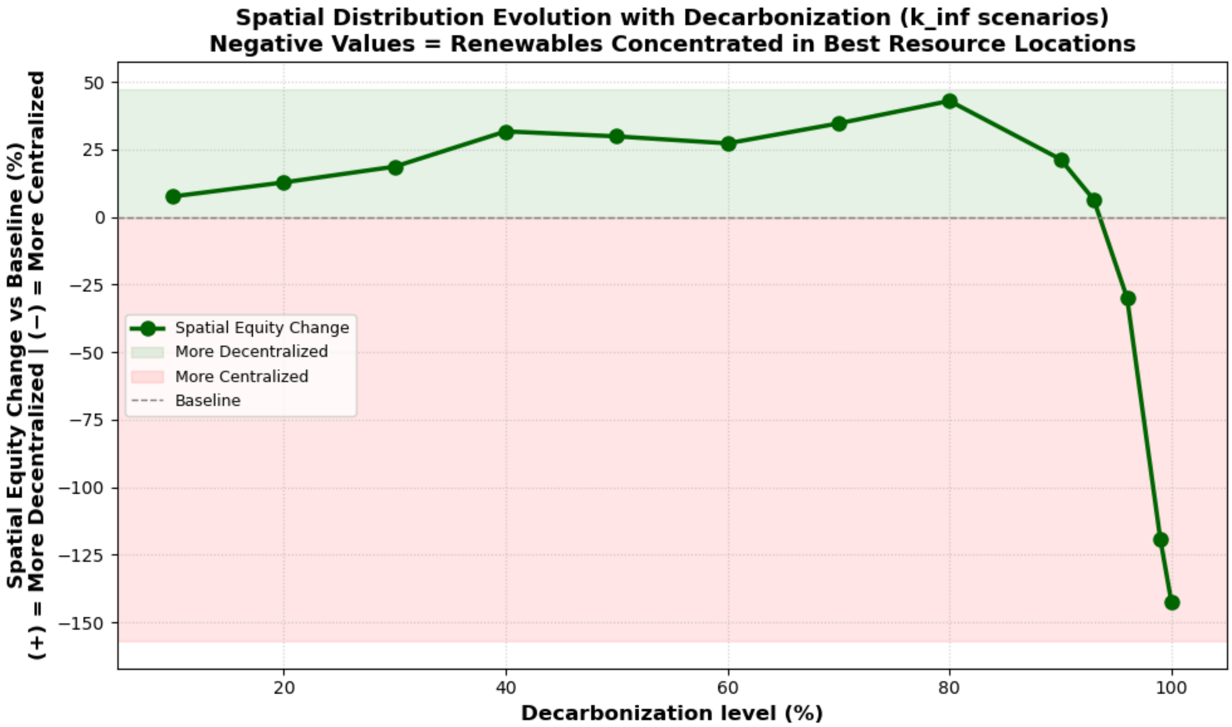


Figure 22: Decarbonized system equity change compared to baseline.

Here it can be seen that as the system decarbonizes the equity initially *increases*, meaning that the system becomes less centralized. However at high levels of decarbonization the equity begins to collapse as the system becomes highly centralized. The expectation would be that since renewable generation buildout is enforced through decarbonization measures, these would be installed in the nodes with the best renewable resources, and that the system would, as a result of this, see a continuos decrease in equity. This is obviously not the case. In order to uncover the underlying mechanism that leads to the initial decentralization when decarbonizing the system, a spatial correlation is defined as seen in equation (26) below:

$$\rho = \frac{\sum_n (C_n^R - \langle C^R \rangle) \cdot (L_n - \langle L \rangle)}{\sqrt{\sum_n (C_n^R - \langle C^R \rangle)^2} \cdot \sqrt{\sum_n (L_n - \langle L \rangle)^2}} \tag{26}$$

Here $C_n^R$ is the installed renewable capacity at node $n$, $L_n$ is the load at node $n$ and $\langle \cdot \rangle$ denotes the mean value across all nodes. This correlation is thus an indicator of how well the installed renewable capacity aligns with the load distribution in the network. A higher positive correlation indicates that the renewable capacity is more evenly distributed across the nodes relative to their load, while a negative correlation indicates a more uneven distribution. Seen below in figure 23 is the evolution of the spatial correlation as the system decarbonizes:

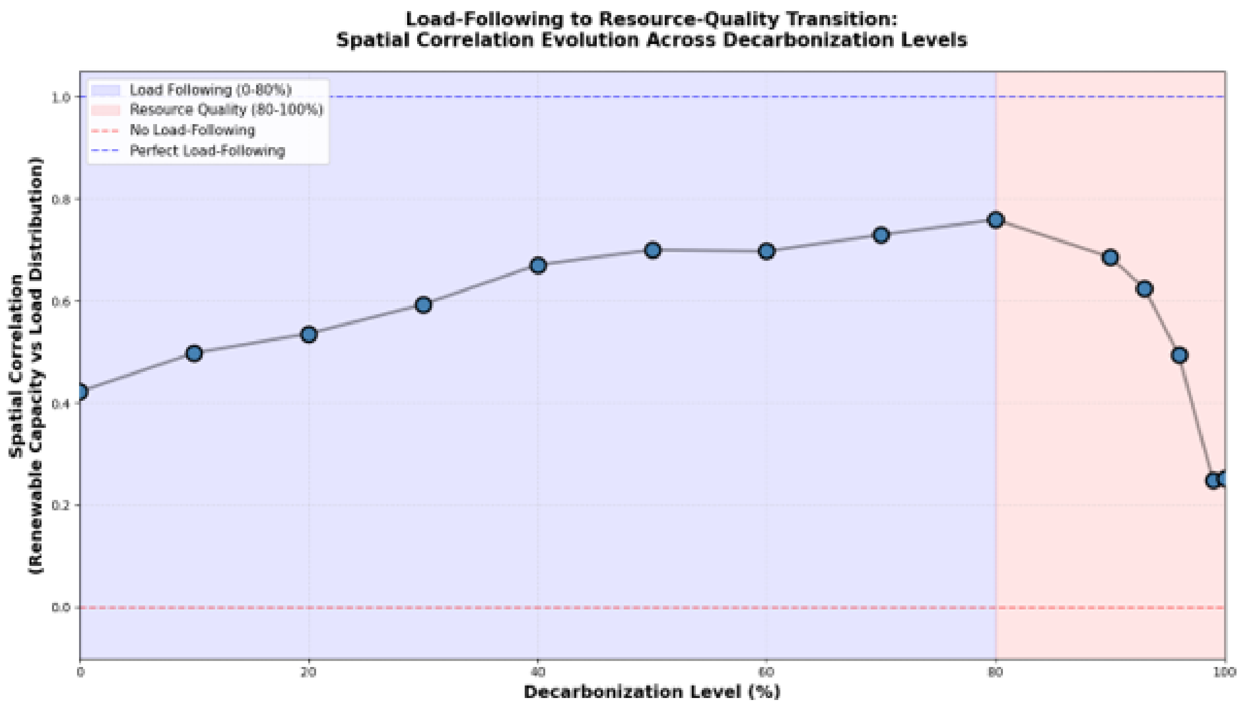


Figure 23: Spatial correlation between renewable capacity and load as the system decarbonizes

As the system decarbonizes the spatial correlation between the load and the installed renewable capacity increases up until 80% decarbonization, where it suddenly decreases to below the original level. This seems to indicate that there are two separate mechanisms, or paradigms, at play here. At lower levels of decarbonization the newly installed renewable generation capacity is following the load centers in the network. When compared to the baseline system this leads to more capacity being installed in more diverse locations, and therefore increasing equity. However as seen in figure 22 above, at very high levels of decarbonization, where renewables dominate the system and backup from dispatchable based generation decreases, the system begins to increasingly optimize for the best renewable resources in order to minimize costs. This centralization of renewable capacity can also be seen directly when observing the capacity maps for the two extremes at 80% and 100% decarbonization seen below in figures 24 and 25 respectively:

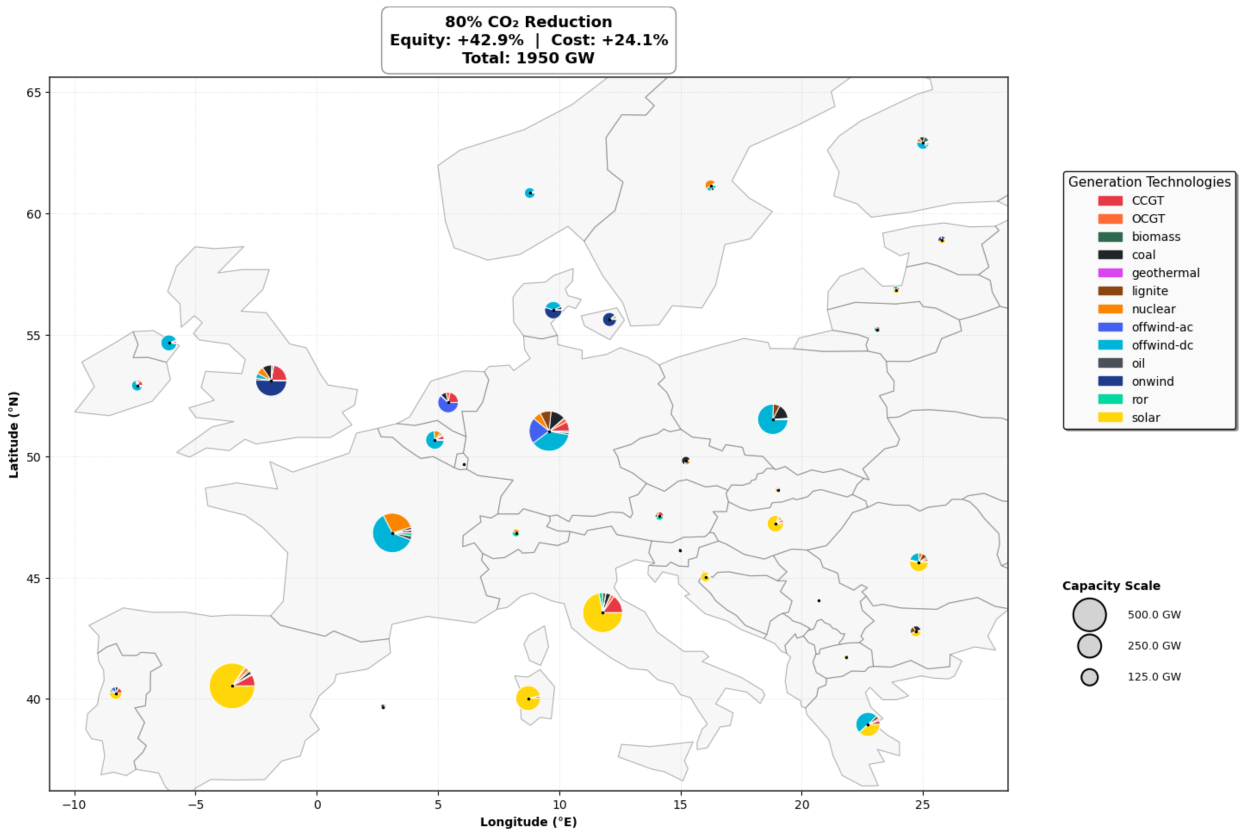


Figure 24: Generation capacity map at 80% decarbonization

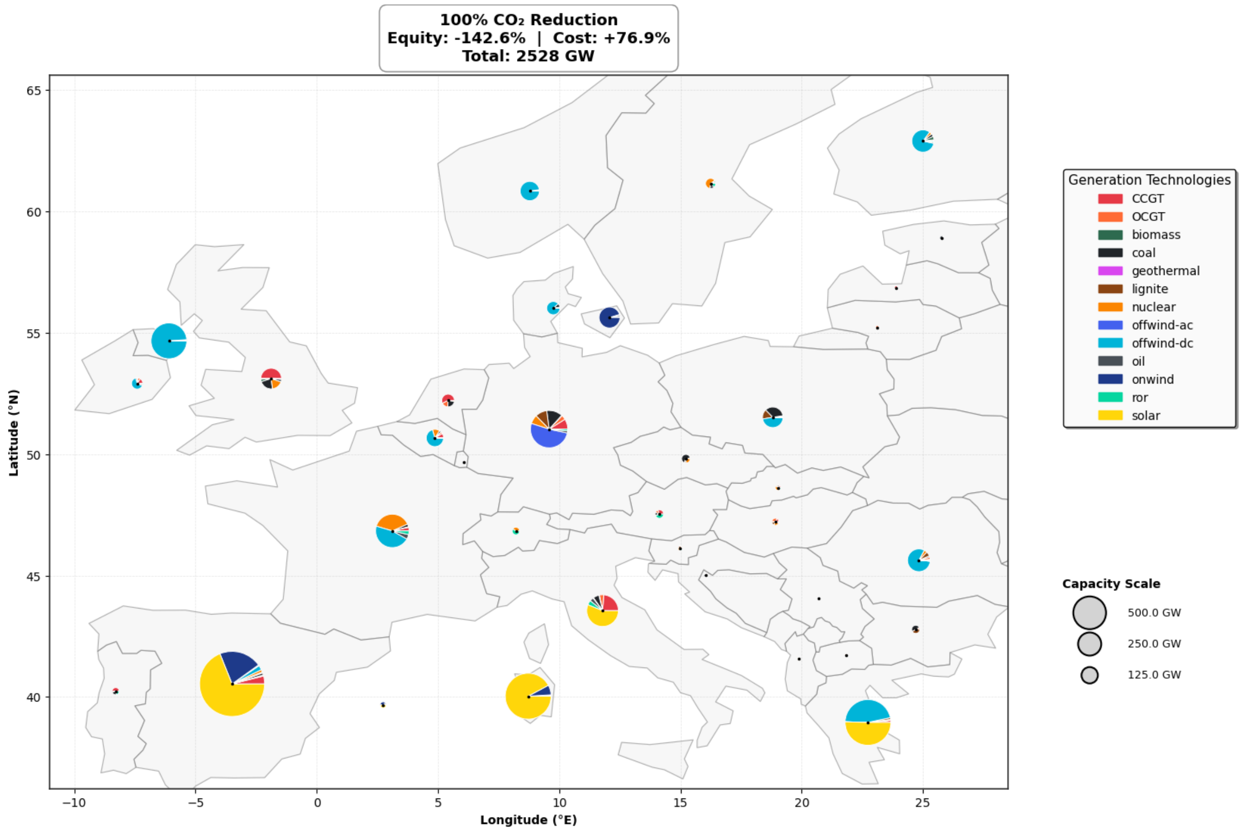


Figure 25: Generation capacity map at 100% decarbonization

When observing regions with good renewable resources such as offshore wind in Northern Ireland and solar in Spain, it can be seen how at 100% decarbonization the favored renewables have been utilized to a much higher degree than at 80%. On the flipside, regions with less favorable renewable resources have seen their installed renewable capacity decrease, like in the northern Balkan region.

### *4.4. Decarbonization mechanisms and interactions*

The decarbonization process, as seen above behaves mostly as one would expect. In general there seems to be a compounding effect in the system, which means that as the system approaches 100% decarbonization, changes between each iteration become more and more pronounced. A pronounced point of transition in paradigm, from load following to resource quality, for many of the metrics shown above, seems to be at 80-90% decarbonization where many of the metrics begin to change more rapidly. The generation mix of electricity by different technologies, favors wind-based generation as these typically have a capacity factor time series (and thus generation) that lines up better with the load time series compared to the solar-based generation. This is caused by the fact that the load pattern in Europe tends to peak during winter, when there are fewer hours of sunlight available to produce electricity. Transmission and storage infrastructure seems to be serviceable up to around 90% decarbonization, after which a massive buildout is required in order to service the load with non-dispatchable renewable generation, which leads to a large increase in system cost. The system also overall becomes more centralized as the system decarbonizes as expected, as the most optimal locations for renewable generation are utilized to a higher, degree in order to meet the decarbonization constraints.

### *4.5. Decentralized systems*

Having performed perturbations of both the level of decarbonization and decentralization constraint K, yields a total of 105 networks. This means that in order to summarize the results, some initial screening is required. For the capacity results this has been done by performing a coefficient of variation (CV) analysis of installed capacity across the different K-constraints for each decarbonization level. This coefficient of variance is defined as $CV = \frac{\sigma}{\delta} \times 100\%$, where $\sigma$ is the standard deviation and $\delta$ is the mean value across the different K-constraints. The results of which can be found in appendix Appendix C: Capacity sensitivity. Unsurprisingly the generation capacity CV analysis, shows that the freely extendable renewable generation technologies solar and wind are impacted the most. Pictured in figure 26 below, is the evolution of the total installed renewable generation capacity for the different levels of decentralization and decarbonization.

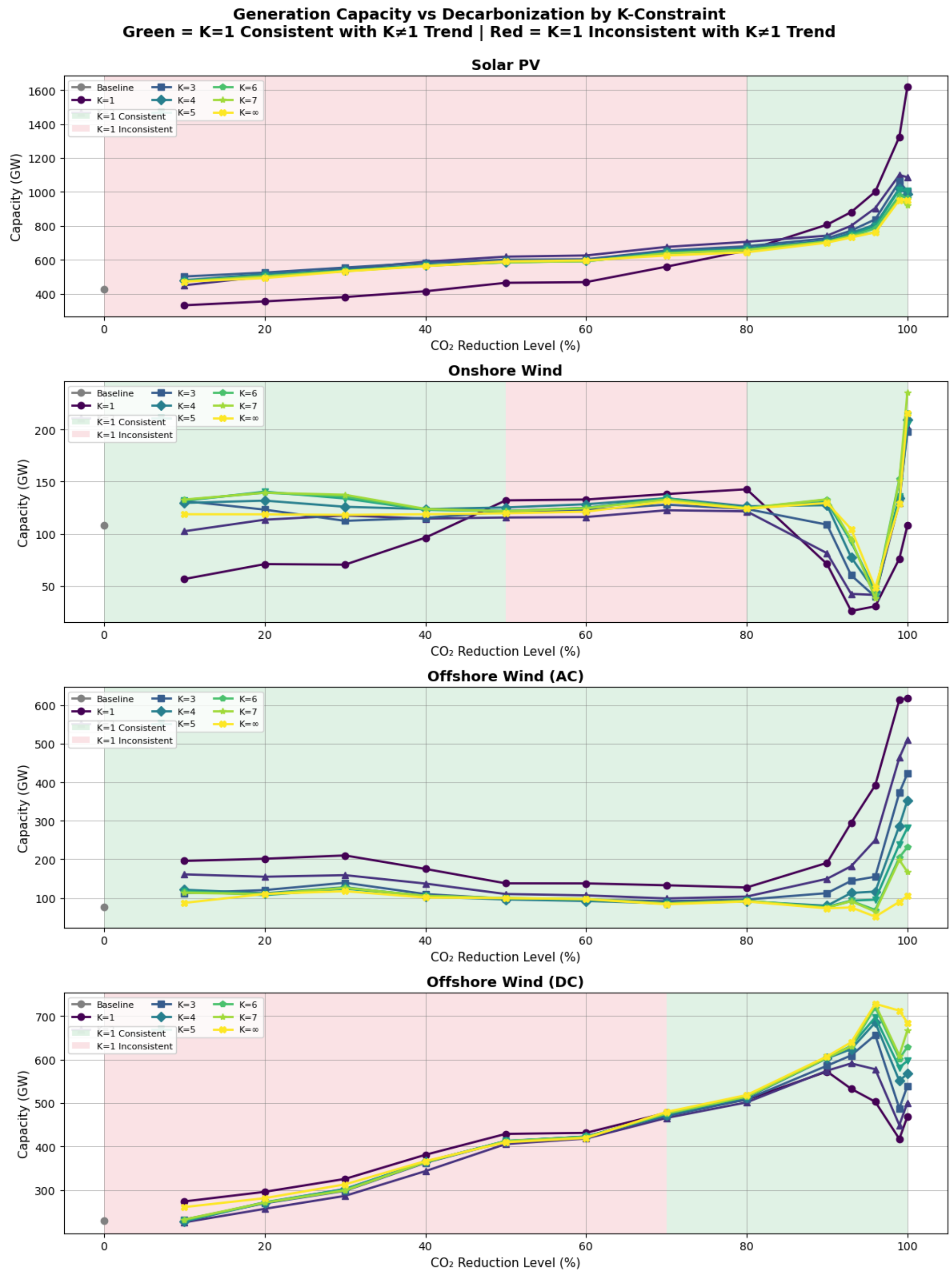


Figure 26: Total installed renewable generation capacity across K-constraints and decarbonization levels.

Interestingly, it can be observed that the extreme scenario of complete decentralization (K=1) is not consistent with the trends of decentralization. Take for example the installed capacity of offshore wind (DC). Here a more decentralized system will in general trend towards a decrease in capacity. However below 70% decarbonization this is not true for $K = 1$ where it actually results in *more* capacity. The color of the background indicates if the capacity for the fully decentralized system (K=1) is consistent with the general trend, when decentralizing or not. This may indicate that the highly restrictive nature of the K=1 constraint leads to rapid changes in optimal decarbonization strategies, depending on the level of decarbonization. The main observations for the four different technologies are listed below:

- **Solar PV:** When decentralizing, the system in general, a more decentralized system leads to an increase in solar capacity, regardless of decarbonization level. However the K=1 scenario diverges from this trend at decarbonization levels below 80% while at higher levels of decarbonization it becomes consistent with the trend.

- **Onshore Wind:** When increasing decentralization, installed capacity decreases. This effect is particularly pronounced at high levels of decarbonization. At low levels of decarbonization the K=1 is consistent, but amplified, before diverging from 50% to 80%, and finally becoming consistent again at higher levels of decarbonization.

- **Offshore Wind (AC):** A more decentralized system leads to more installed capacity across all levels of decarbonization, with a significant increase at low levels of decarbonization, and a huge increase at high levels of decarbonization. The extreme K=1 scenario is consistent with this trend, at all levels of decarbonization.

- **Offshore Wind (DC):** In general a decentralizing system will lead to a decrease in installed capacity, particularly at high levels of decarbonization. At low levels of decarbonization (>70%) the K=1 scenario is inconsistent with this trend but at high levels of decarbonization it becomes consistent.

Overall it seems that decentralization leads to an increase in solar and AC-offshore wind capacity, as these can that can be more easily distributed across the network, compared to onshore wind farms and DC-offshore wind. In order to better understand the underlying mechanisms and why the K=1 scenario behaves differently from the other decentralization levels, further analysis is required. Take for example the mean of the capacity factor time series, for solar PV, at different levels of decentralization and decarbonization seen below in figure 27:

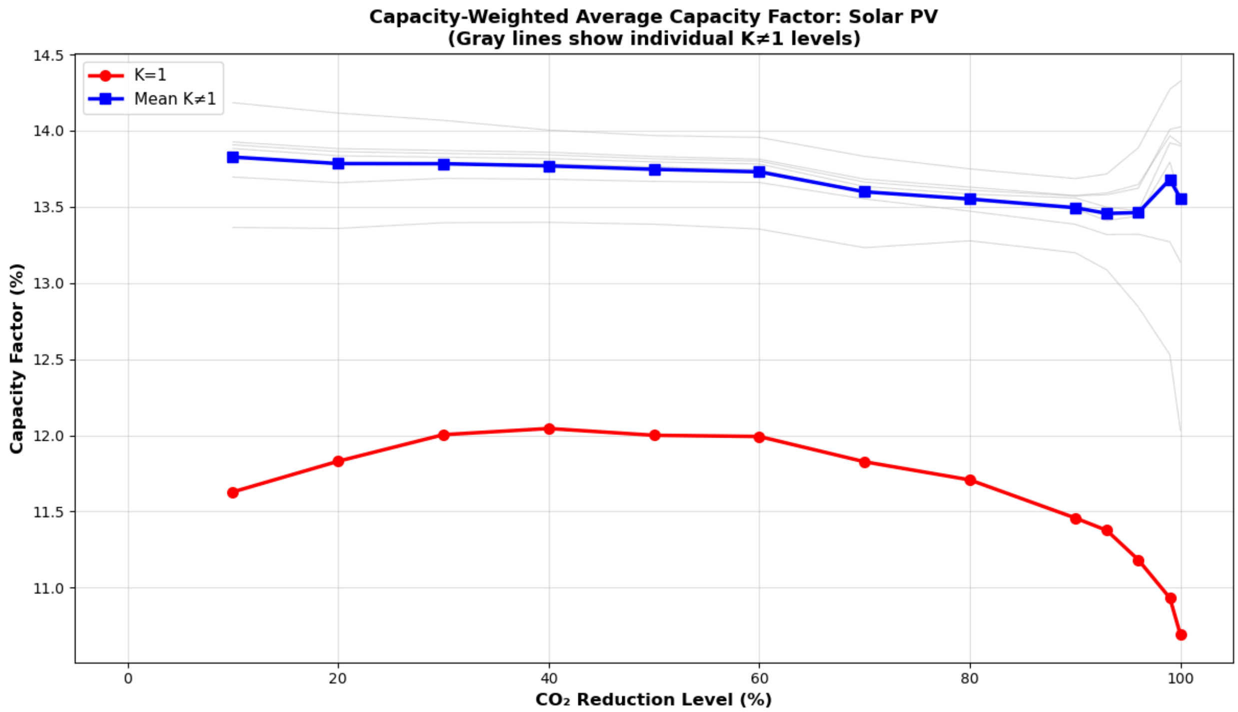


Figure 27: Mean capacity factor for solar PV across K-constraints and decarbonization levels.

Here it is evident that, as the system decarbonizes, for decentralization levels $K \neq 1$ the mean capacity factor for solar remains fairly consistent at all levels of decarbonization. However for the extreme K=1 scenario the mean capacity factor decreases significantly as the system decarbonizes and is in general 13%-21% lower than the mean of the other decentralization levels. This degradation of resource quality also appears for the other technologies, albeit not as large. The plots of capacity factors can be found in the supplementary material under Appendix Appendix F: Mean capacity factors, where it can be seen that for $K = 1$ the poorest performing capacity factors belong to solar PV, followed by Onshore wind, then Offshore AC and finally Offshore DC. The decrease in capacity factor is caused by the fact that decentralizing forces renewable capacity buildout in worse locations, for all four technologies at all levels of decarbonization. This effect is particularly pronounced at K=1. This then explains why at some levels of decarbonization, as seen in figure 26, the extremely decentralized $K = 1$ leads to a decrease in installed capacity due to poor performance. However it does not explain why at some levels of decarbonization $K = 1$ *increases* capacity of the very same poorly performing technologies.

At lower levels of decarbonization the $K = 1$ scenario essentially says: *Meet a modest $CO_2$ constraint at minimum cost, while spreading extra renewables proportionally across all nodes (bad solar and bad wind alike).* This means that in order to keep costs low the optimizer installs additional renewable capacity which produces the most MWh per forced system-wide GW, and has a low degree of curtailment. This punishes solar PV capacity, as solar has the largest decrease in capacity factor when decentralizing as seen above. The optimizer therefore ends up suppressing solar capacity buildout at low levels of decarbonization. This trend however reverses at high levels of decarbonization. Here the question changes from *How do we meet a modest $CO_2$ constraint at minimum cost while spreading renewables proportionally across all nodes?* to *How do i meet the load with all renewable energy despite correlation and curtailment?*. Here, as the system is forced to decarbonize completely, the increase in necessary wind capacity to meet the load leads to very large amounts of curtailment, as wind generation is highly correlated across the network. at 100% decarbonization and $K = 1$ the wind curtailment reaches 12%-16% across all wind technologies, while solar remains around 2% curtailment. This seems to indicate that wind deployment has reached saturation in the system. Solar PV capacity then becomes the "escape valve" as solar generation is better suited for storage and balancing with the prevalent technology (batteries) and produces energy in different hours than wind. This forced diversification then leads to a massive solar buildout, at very high levels of decarbonization, despite the poor performance relative to wind.

However, the most important takeaway from this discussion is the fact that the extreme decentralization scenario of K=1 behaves fundamentally different from the other decentralization levels, due to the highly restrictive nature of the constraint and the system wide heterogeneity becoming an efficiency bottleneck. This happens as $K = 1$ represents a fundamentally different planning paradigm. Here the problem changes from a spatial and technological optimization, to a purely technological optimization. What follows is then a system where every additional GW capacity must be replicated system wide, due to the decentralization constraint. This explains the sudden changes in decarbonization strategy and leads to changes in dominating technologies, when compared to the other decentralization levels, even at the same level of decarbonization.

It is known that, due to the variance in capacity factor time-series for different generating technologies, the installed capacity and the annual generation do generally follow the same trends and are therefor not depicted here. This has been confirmed by reviewing the generation results across the different K-constraints and decarbonization levels, which can be found in the supplementary material under Appendix Appendix D: Annual generation evolution for decentralization and decarbonization. As for the transmission capacity results, these can be seen in Appendix I: Figures for decentralization evolution in figure 35:

Unsurprisingly, as the system is forced to decentralize, the total transmission capacity decreases, a trend that is increasingly pronounced at low and high levels of decarbonization, where the system is known to already be increasingly centralized, leading to a higher sensitivity to decentralization measures. Transmission of course works in congruence with storage as part of the nodal balancing. In order to analyze the interactions between the different types of storage capacity, the decentralization measure, and the level of decarbonization, initially a coefficient of variance analysis was performed in order to determine which storage technologies are the most sensitive to the decentralization process. The results of this analysis can be found in Appendix Appendix E: Coefficient of Variance analysis for Storage capacity. Here it can be seen that the maximum capacity limited nature of the water-based storage technologies, means that these are not sensitive to decentralization measures. While hydrogen based storage is sensitive to decentralization, overall installed capacity remains negligible. Pictured below in Appendix I: Figures for decentralization evolution figure 36 is the installed storage capacities of batteries for all levels of decarbonization and decentralization. Here it can be seen that, as the system decarbonizes and decentralizes the installed battery capacity initially decreases a little due to the load following mechanism discussed above before increasing dramatically at high levels of decarbonization and decentralization. This aligns well with the idea that as the system is forced to decentralize the need for local nodal balancing increases. It also aligns with the fact that the decrease in transmission capacity when decentralizing means that less remote balancing can be performed through the transmission network and as such more localized storage is needed.

Regarding the economic metrics of the decentralized system, the mean marginal demand-weighted price evolution and the total scarcity rent evolution can be seen below in Appendix I: Figures for decentralization evolution figure 37 and 38 respectively. As expected the mean marginal demand weighted price follows the same trend as seen in the decarbonizing system above. The price increase when enforcing decentralization is also as expected, due to the fact that as the system is forced to decentralize, it can no longer fully utilize the best renewable resources and must instead rely on more expensive local resources. This can also be seen in the fact that at very high levels of decarbonization, the price increase when decentralizing is more pronounced, as the system is more reliant on renewable generation and is solidly within the resource–quality paradigm. As for the evolution of the total scarcity rent across the different K-constraints, and decarbonization level, scarcity rents also interact with decentralization as the system becomes increasingly constrained, which develops scarcity rents as explained previously. Total system cost behaves in an intuitive manner, with an increase in decentralization leading to a higher total system cost, especially at high levels of decarbonization where the resource quality paradigm is in full effect. A cost increase when decentralizing, is to be expected as the system, especially at higher levels of decarbonization are heavily reliant on generation capacity installed in nodes with good renewable resources. Forcing decentralization then means that the system can no longer fully utilize these best resources, and must instead rely on less efficient and more expensive local resources.

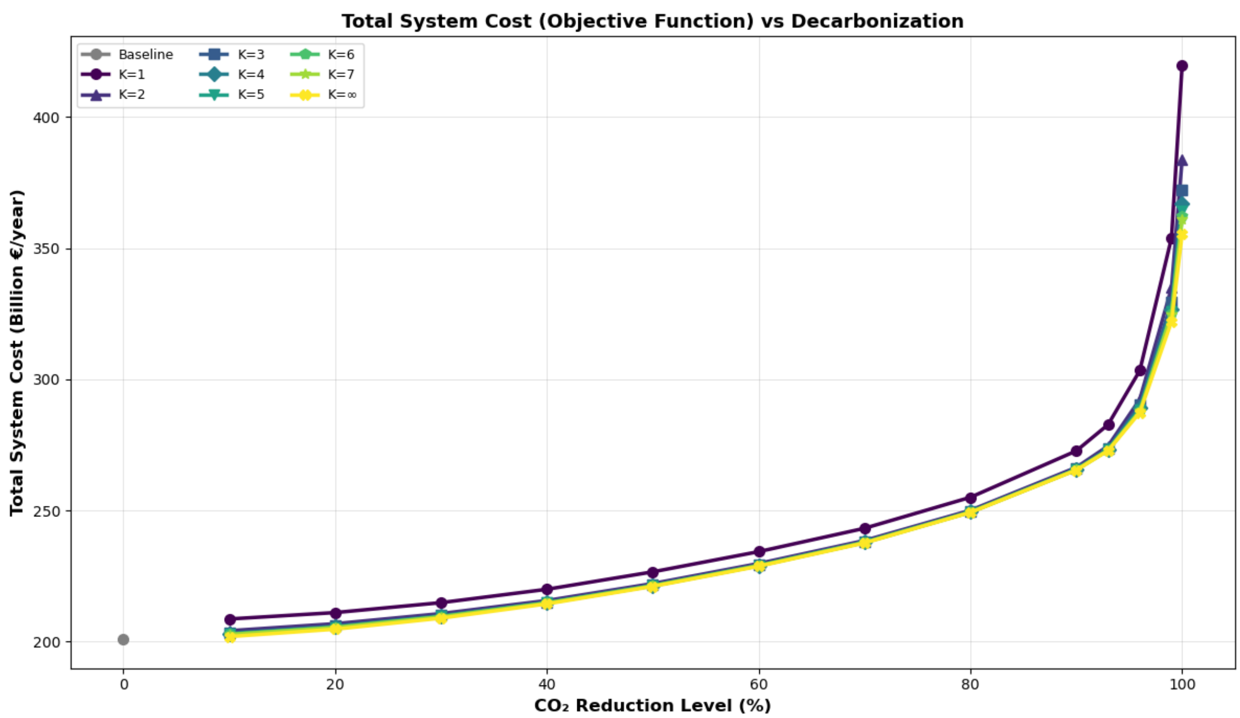


Figure 28: Total system cost across K-constraints and decarbonization levels.

Finally it is of great interest to observe how the SDRCL behaves when both decentralization and decarbonization are enforced. The reason for the interest is the fact that the decentralization measure is applied as a variation of the decarbonized systems and the K-constraint method of achieving decentralization may therefore have varying degrees of success decentralizing the system depending on the level of decarbonization. Pictured below in figure 29 is the evolution of the equity measure across the different K-constraints and decarbonization levels:

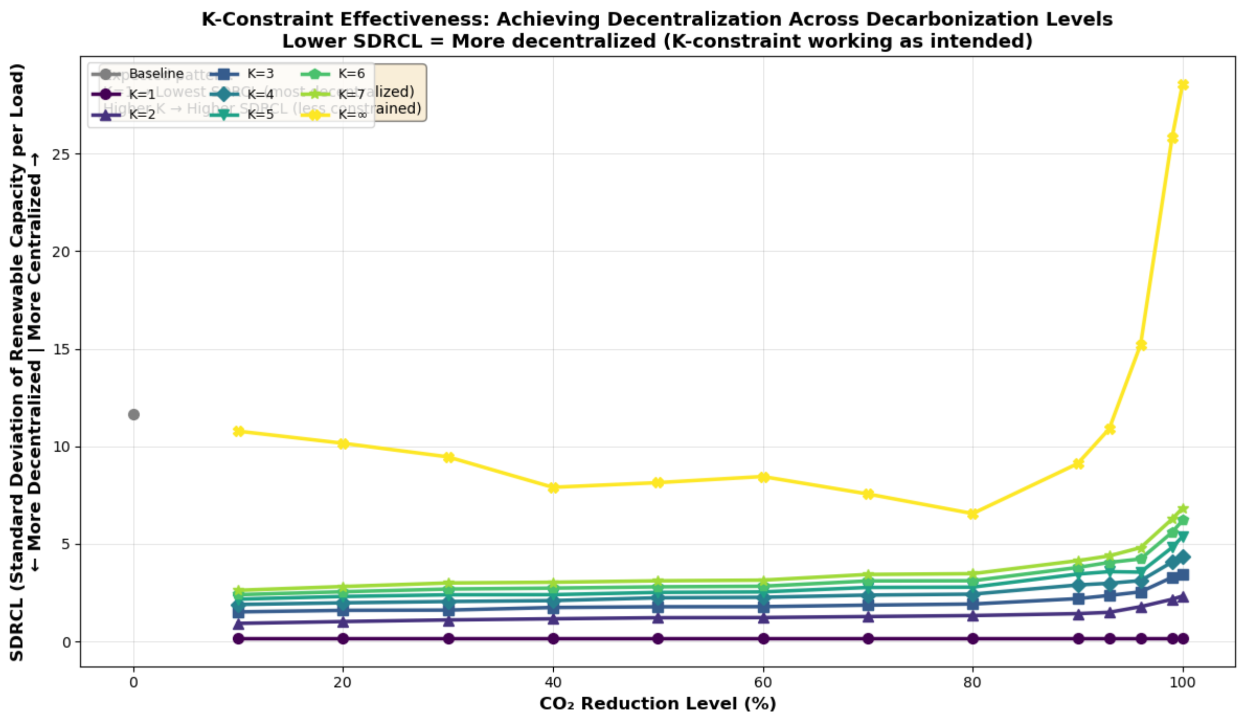


Figure 29: Equity measure across K-constraints and decarbonization levels.

As evident from the graph the k-constraint measure has successfully decentralized the system across all levels of decarbonization. Interestingly as seen in figure 29 above it appears that the decentralization constraint tightens, the system becomes less sensitive to the decarbonization level. The main takeaway from this figure however, is that the K-constrained networks all achieve significant reduction in centralization, while having similar cost premiums as seen in figure 28 above. This would indicate that the solution space in regards to cost optimizing while reducing centralization is actually quite flat. This is a very significant result and might indicate that decentralizing systems is not as cost prohibitive as previously thought. A full breakdown of cost and equity tradeoff can be found in Appendix G: Cost-Equity tradeoff. The best performing constraint ($K = 7$) is seen for a few select levels of decarbonization below in table 6 below which summarizes the cost-equity trade-off between the decentralization scenario of K=7 and the extreme decentralization scenario of K=1, as well as the scenario not subject to decentralization constraints ($K = \infty$):

Table 6: Cost-Equity Trade-off Analysis: Optimal Decentralization (K=7) vs Complete Decentralization (K=1)

| **Decarb.** | **Cost Premium (%)** | | **Equity Benefit (%)** | | **K=7 Relative to K=1** | | **Efficiency vs K=$\infty$** | |
|---|---|---|---|---|---|---|---|---|
| | K=1 | K=7 | K=1 | K=7 | Cost | Equity | K=1 | K=7 |
| 30% | 2.82 | 0.23 | 98.3 | 68.3 | 8.2% | 69.5% | 34.8x | 296.0x |
| 50% | 2.51 | 0.09 | 98.1 | 61.8 | 3.4% | 63.0% | 39.1x | 716.0x |
| 100% | 18.11 | 1.56 | 99.4 | 76.1 | 8.6% | 76.5% | 5.5x | 48.7x |

Here it can be seen that $K = 7$ seems to capture a very large percentage of the equity gains of $K = 1$ but for a fraction of the cost premium. Critically, moderate decentralization (K=7) achieves 62-76% of K=1's equity benefits at only 3-9% of the cost penalty across all decarbonization levels, demonstrating a 7-34$\times$ superior cost-equity trade-off. This suggests partial decentralization is far more economically viable than complete spatial equity. In order to better illustrate how the decentralization measure has impacted the equity of the system, Pictured below is the distribution of the renewable capacity to load ratio used to calculate the SDRCL for 100% decarbonization for $K = \infty$ (free to centralize) and $K = 7 \wedge K = 1$ constrained scenarios.

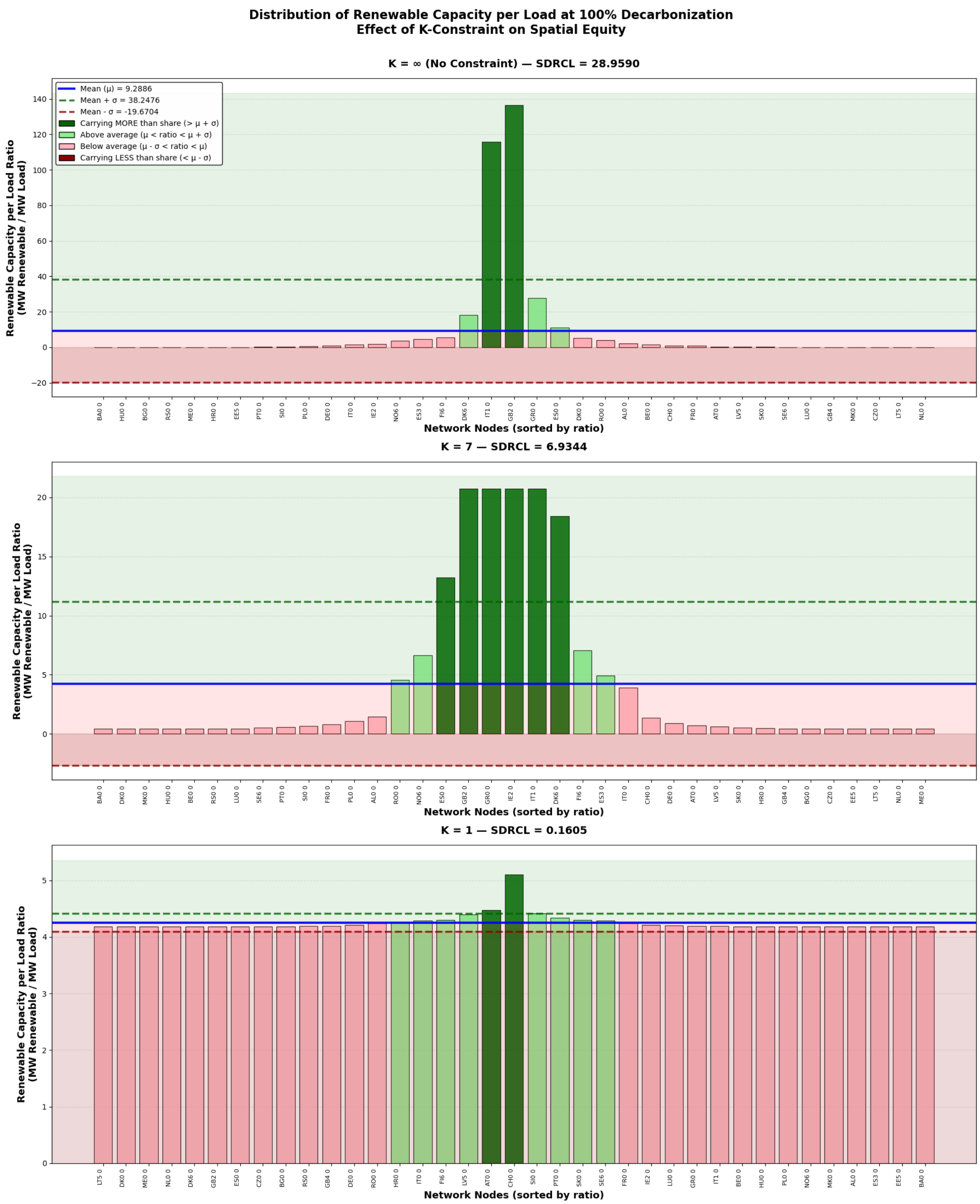




Figure 30: Distribution of renewable capacity to load ratio at 100% decarbonization for different K-constraints.

Here it is clearly illustrated that as K approaches 1 the distribution is forced to "flatten" meaning that all nodes must have similar renewable capacity to load ratios. However even at $K = 1$ there is still a little variance in the distribution as this is necessary in order to meet the load with the available renewable resources and create a solvable optimization problem. The statistics for the distributions can be found in Appendix H: Distribution statistics.

### *4.6. Decentralization mechanisms and interactions*

While there seems to be a significant overlap between the mechanisms governing the network structure of the decarbonized and decentralized systems, some interactions change key characteristics depending on the level of decarbonization. These changes are however mostly to the extreme decentralization scenario of $K = 1$. While the key transition region of the decarbonization sweep at around 80-90% decarbonization is still present, the underlying mechanisms driving the decarbonization strategy change depending on the level of decarbonization.

A secondary transition from paradigm to paradigm occurs when decentralizing the system with total equity (K=1). These two paradigms can be summarized as follows:

- **Low to moderate decarbonization (0-80%):** Here the system is primarily driven by the load following mechanism where renewable capacity is installed in order to meet the decarbonization constraint while following the load centers in the network. This leads to a more decentralized system as renewable capacity is installed in more diverse locations.

- **High to total decarbonization (80-100%):** Here the system is primarily driven by the resource quality mechanism where renewable capacity is installed in order to meet the decarbonization constraint while prioritizing the best renewable resources in the network. This leads to a more centralized system as renewable capacity is installed in the locations with the best renewable resources

- **Moderate decentralization ($K_{\mathbf{inf}}$ to $K = 2$):** Here the system is forced to decentralize which maintains the interactions as seen from the decarbonization but tends to become increasingly extreme at high levels of decarbonization.

- **Extreme decentralization (K=1):** Here the system is forced to decentralize completely regardless of the level of decarbonization. This leads to unique decarbonization strategies when compared to the other decentralization levels.

## 5. Discussion

Having thoroughly reviewed and analyzed the results from the workflow it is now important to consider the results and trends in a broader context. In real-world applications, energy system modelling does not stand alone but is a tool to be used to make informed decisions and guide policy makers. It is therefore of utmost importance to take a step back and review the results of this master's thesis in such a broader context. Consider the first two research questions posed in the introduction:

- How, if at all, does enforcing decentralization change the cost-optimal pathway to decarbonization in a European energy system?
- Which system mechanisms are responsible for the interactions between decarbonization and decentralization?

As evident from the results the answer to these two questions are intrinsically linked. From a system modelling perspective it is clear that when performing a simple least-cost optimization of a decarbonizing system, this comes with inherent political assumptions. The heavily centralized nature of the least-cost systems, especially at higher levels of decarbonization, may not always align with political paradigms and goals. On the other hand decentralization measures as enforced through equity constraints, while more palatable for politicians, fundamentally change the underlying mechanisms of the decarbonization process. This then in turn leads to alternative transition pathways and comes with a cost premium. The cost of decentralization is also highly dependent on the level of decarbonization and the degree of decentralization desired. This then further complicates the decision making process for politicians and system planners as they must balance the cost of decentralization with the political benefits.

Understanding the interacting paradigms that govern these processes, is key to understanding how the transition pathways are reshaped, and its impact on policy making. Firstly, when considering the *decarbonizing* system, this is governed by two main planning-paradigms: The load-following paradigm and the resource-quality paradigm. Given the current state of the European energy system, it seems that the transition towards a decarbonized system will initially, in a collaborative least-cost system, increase the equity in the system relative to the baseline. This occurs as renewable capacity will see substantial buildout in countries with large shares of the european load, that may not up until now have invested significantly in renewable generation. However at a certain inflection point, estimated to be around 80-90% decarbonization, the system will transition from one planning paradigm (load following) to another (resource quality). Here it becomes necessary in order to service the load and keep costs as low as possible to begin to centralize renewable capacity in a few regions with the best renewable resources. This could for example be Spain having a larger share of solar capacity buildout than their share of the european load, and the UK and Ireland having a larger share of offshore wind capacity buildout than their share of the european load. Thus, when removing the last remnants of fossil fuel generation from the system the system will become increasingly centralized. This also causes a large buildout of transmission and storage infrastructure needed to balance the highly renewable system and as such leads to large cost increases. This is supported by the analysis of the infra-marginal rent which reveals how system bottlenecks evolve during the decarbonization process. At low levels of decarbonization the scarcity rents are driven by fossil fuel generation capacity while at high levels of decarbonization, when the paradigm shifts, the scarcity rents are driven by renewable generation capacity, transmission capacity and storage capacity. The shift in scarcity rent drivers further supports the idea of two separate planning-paradigms governing the decarbonization process and shows the fundamental change in which system constraints are dominant.

If a system planner, or politician, then wishes to combat this natural centralization that stems from the resource-quality paradigm, say for example due to security concerns or political reasons, then *decentralization* measures can be enforced. It is, however, evident from the results that it is very important that this is done with care. As described in the methodology of this thesis' if decentralization is enforced as a hard limit and not normalized by some other system measure, it can lead to the decentralization constraint becoming the dominant limit in the system instead of the decarbonization constraint. This then leads to the system overshooting decarbonization targets and large cost premiums. The work done in this thesis' has shown that decentralization can be successfully enforced through the K-constraint method while still allowing the system to decarbonize in a least-cost manner. This process, like that of decentralization, is also governed by two main planning-paradigms: One for increasing equity at low to moderate levels of decentralization ($K \neq 1$), and one for extreme decentralization ($K = 1$). For both paradigms the system can successfully increase equity by a significant amount for relatively low cost premiums. This is extremely important as it indicates that decentralization is not as cost prohibitive as previously thought. However the two planning-paradigms, while achieving similar results, differ wildly in their underlying mechanisms and strategies.

At low to moderate levels of decentralization the system primarily follows the same mechanisms as the decarbonizing system ,although somewhat amplified, where at low levels of decarbonization the load-following mechanism dominates and at high levels of decarbonization the resource-quality mechanism dominates.

However at extreme decentralization the system is forced to decentralize completely regardless of the level of decarbonization. This leads to unique decarbonization strategies when compared to the other decentralization levels. Here at low levels of decarbonization the system primarily prioritizes cost minimization while spreading renewable capacity proportionally across all nodes, leading to suppression of solar capacity buildout. This happens due to the extreme decentralization paradigm forcing renewable deployment in suboptimal locations. This then in return lowers the mean capacity factor of solar especially. The degradation if capacity factor is the root cause of the extreme decentralization paradigms unique behavior at low levels of decarbonization. At high levels of decarbonization the system primarily prioritizes meeting the load with all renewable energy. This means that the level of curtailment for wind becomes very high, while solar curtailment remains low. This is a symptom of the saturation of wind generation in the system and leads to solar capacity becoming the escape valve as additional wind deployment would simply lead to more curtailment due to the high correlation of wind generation across the network. This forced diversification then leads to a massive solar buildout at very high levels of decarbonization despite the poor capacity factor. In addition the extreme decentralization paradigm leads to large increases in battery storage capacity as the system is forced to balance more locally due to the decrease in transmission capacity which constitutes a large cost increase.

As for the thrid research question posed in the introduction:

- What sacrifices must be made in terms of cost and system design in order to achieve a decentralized energy system?

It is then evident from the results, that with the planning paradigms in mind, a tradeoff between decentralization and cost must be made. It is however critical that this does not come at the expense of decarbonization goals. The results from this thesis' indicate that decentralization can be successfully enforced, although at a cost premium ranging from 0.39% at low levels of decarbonization and decentralization to 18.11% at high levels of decarbonization and decentralization. This is an extremely important result as it indicates that decentralization and decarbonization are not mutually exclusive goals and can be achieved together. Another sacrifice that must be made when enforcing decentralization is the degradation of capacity factor for renewable generation technologies as the system is forced to install capacity in suboptimal locations. This then leads to an increase in installed capacity and the cost premiums seen above. Finally decentralization also leads to the system becoming increasingly reliant on energy storage in order to balance the system locally which again leads to increased costs.

While overall the results from this thesis' indicate that both decarbonization and decentralization can be successfully achieved together, the volatility of the extreme decentralization paradigm indicates that this approach may not be suited for real-world applications. The highly restrictive nature of the extreme decentralization constraint means that small changes in the system can lead to large changes in the decarbonization strategy. This could lead to unforeseen consequences and instability in the energy system transition if applied in real-world scenarios. economically the extreme decentralization paradigm also lacks support as the cost of spatial equity varies wildly depending on the level of decarbonization. Therefore it is recommended that if decentralization measures are to be enforced in real-world applications, that this is done through low to moderate levels of decentralization where the system can still follow least-cost decarbonization strategies with more predictable cost premiums. This cost premium is however non-linear with the level of decentralization and again means that careful consideration must be taken when choosing the level of decentralization to enforce.

In regards to the limitations of the work done for this master's thesis the major point of improvement is the cost modelling of hydrogen storage as the conservative estimates of efficiency from the DEA seemingly excludes hydrogen storage from becoming viable in the system. Further research into more accurate cost estimates for hydrogen storage would be of great interest as hydrogen storage is often described as a very important part of the puzzle for balancing highly renewable energy systems. The model itself is also simplified by the fact that it uses a single weather year, which may not capture the full variability of renewable resources. The network model is limited to electricity and contains no sector coupling. AS sectors in Europe become electrified this may have significant impacts on the decarbonization and decentralization pathways. Finally, on the political side of the issue no direct behavioral or political acceptance modeling has been included in the model with total load-based equity as a proxy for political acceptance. Including more detailed political and behavioral models could significantly impact the results and would be of great interest for future research, as load based equity may, by chance, disadvantage some regions while artificially inflating other regions installed capacity.

Although the results and analysis presented in this thesis' is comprehensive there are of course many different avenues of research that could be pursued. In order to do so it would be of interest to consider some of the more novel approaches developed for this thesis. Firstly the K-constraint method for enforcing decentralization in the system without over-decarbonizing seems to be a promising and efficient solution to modelling decentralization. Having previously worked with the Gini coefficient as a measure of decentralization and using it to define decentralization constraints, this K-approach seems more intuitive and easier to implement. Additionally the use of the SDRCL as an equity measure seems promising as it is easy to calculate and understand while providing crucial insight into the system distribution of renewable capacity relative to load. Further research into these two methods and their applications in different contexts would be of great interest. Furthermore the decarbonization paradigms of load-following and resource-quality would be interesting topics to investigate further in order to uncover if these are fundamental mechanisms governing the decarbonization process in general or if they are specific to this model and context. The role of the extreme decentralization paradigm and its fundamental differences from the moderate decentralization paradigm would also be an interesting topic to investigate further.

As briefly mentioned using alternative equity measures and decentralization constraints would also be of great interest. For example by modelling political acceptance in a number of different ways their impacts on the decarbonization pathways may be monitored. Finally including sub-national levels of decentralization and increasing the spatial resolution of the model would also be of great interest as decentralization could be explored as a local phenomenon.

## 6. Conclusion

Full decarbonization increased total system cost by approximately 80% relative to the baseline, while total installed generation capacity increased by roughly 78%. A pronounced paradigm transition was observed between 80–90% emission reduction, as shown by a rapid growth in storage and transmission requirements as well as a heavy centralizing effect. Without decentralization constraints, system equity initially improves but collapses at high decarbonization levels due to concentration in regions with optimal renewable resources. The cost of spatial equity varies dramatically with decarbonization ambition. At 30-50% decarbonization, K=1 imposes modest cost premiums (2.5-2.8%) as distributed renewables can still meet most requirements. However, at 100% decarbonization, K=1 cost premiums reach 18% due to forced solar over-building and capacity factor degradation. Moderate decentralization (K=7) achieves 62-76% of K=1's equity benefits at only 3-9% of the cost penalty across all decarbonization levels, demonstrating a 7-34× superior cost-equity trade-off. This suggests partial decentralization is far more economically viable than complete spatial equity. Two distinct decentralization paradigms were identified: moderate decentralization follows decarbonization-driven load-following and resource-quality mechanisms, while extreme decentralization (K=1) enforces uniform capacity distribution, leading to unique strategies and higher costs. Overall, this thesis contributes a linear and transparent framework for enforcing decentralization in macro energy system model optimization and enables exploration of equity-cost tradeoffs. It demonstrates the fact that decarbonization and decentralization can be achieved together, although at a cost premium, and highlights the importance of carefully selecting decentralization levels to balance equity goals with economic viability.

## 7. Acknowledgments

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**Kiri Kristine Aden-Dam**

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**Agnete Marie Hougaard**

For at tænde nok bål.

## 8. References

### References

[1] The Paris Agreement | UNFCCC.
URL https://unfccc.int/process-and-meetings/the-paris-agreement

[2] R. A. Rodriguez, S. Becker, M. Greiner, Cost-optimal design of a simplified, highly renewable pan-European electricity system 83 658–668. doi:10.1016/j.energy.2015.02.066.
URL https://www.sciencedirect.com/science/article/pii/S0360544215002212

[3] S. Pfenninger, A. Hawkes, J. Keirstead, Energy systems modeling for twenty-first century energy challenges 33 74–86. doi:10.1016/j.rser.2014.02.003.
URL https://www.sciencedirect.com/science/article/pii/S1364032114000872

[4] H. Henke, M. Dekker, F. Lombardi, R. Pietzcker, P. Fragkos, B. Zakeri, R. Rodrigues, J. Sitarz, J. Emmerling, A. Fattahi, F. Dalla Longa, I. Tatarewicz, T. Fotiou, M. Lewarski, D. Huppmann, K. Kavvadias, B. Zwaan, W. Usher, Comparing energy system optimization models and integrated assessment models: Relevance for energy policy advice 3 69. doi:10.12688/openreseurope.15590.2.

[5] E. H. Eriksen, L. J. Schwenk-Nebbe, B. Tranberg, T. Brown, M. Greiner, Optimal heterogeneity in a simplified highly renewable European electricity system 133 913–928. doi:10.1016/j.energy.2017.05.170.
URL https://www.sciencedirect.com/science/article/pii/S0360544217309593

[6] The Scale of Russian Sabotage Operations Against Europe's Critical Infrastructure.
URL https://www.iiss.org/research-paper/2025/08/the-scale-of-russian--sabotage-operations--against-europes-critical--infrastructure/

[7] GNU General Public License version 2.
URL https://opensource.org/license/gpl-2-0/

[8] Snakemake | Snakemake 9.14.0 documentation.
URL https://snakemake.readthedocs.io/en/stable/

[9] K. Hedegaard Aden, Thesis-workflow for 'Fairness in a decarbonizing Europe'.
URL https://github.com/KrisAden/Thesis-workflow

[10] GNU General Public License v2.0 - GNU Project - Free Software Foundation.
URL https://www.gnu.org/licenses/old-licenses/gpl-2.0.en.html

[11] T. Brown, J. Hörsch, D. Schlachtberger, PyPSA: Python for Power System Analysis, Journal of Open Research Software 6 (4) (2018). arXiv:1707.09913, doi:10.5334/jors.188.
URL https://doi.org/10.5334/jors.188

[12] J. Hoersch, F. Hofmann, D. Schlachtberger, T. Brown, Pypsa-eur: An open optimisation model of the european transmission system, Energy Strategy

Reviews 22 (2018) 207–215. arXiv:1806.01613, doi:10.1016/j.esr.2018.08.012.

[13] J. Hörsch, F. Neumann, F. Hofmann, D. Schlachtberger, M. Frysztacki, J. Hampp, P. Glaum, T. Brown, PyPSA-Eur: An Open Optimisation Model of the European Transmission System (Dataset). doi:10.5281/zenodo.7646728.
URL https://zenodo.org/records/7646728

[14] B. U. Schyska, A. Kies, M. Schlott, L. von Bremen, W. Medjroubi, The sensitivity of power system expansion models 5 (10) 2606–2624. doi:10.1016/j.joule.2021.07.017.
URL https://www.sciencedirect.com/science/article/pii/S254243512100355X

[15] Transparency Platform.
URL https://transparency.entsoe.eu/load/total/dayAhead?appState=%7B%22sa%22%3A%5B%22CTY%7C10YBE----------2%22%5D%2C%22st%22%3A%22CTY%22%2C%22mm%22%3Atrue%2C%22ma%22%3Afalse%2C%22sp%22%3A%22HALF%22%2C%22dt%22%3A%22TABLE%22%2C%22df%22%3A%5B%222025-12-01%22%2C%222025-12-01%22%5D%2C%22tz%22%3A%22CET%22%7D

[16] Carbon Free Europe.
URL https://www.carbonfreeeurope.org

[17] Technology Data for Generation of Electricity and District Heating | Energistyrelsen.
URL https://ens.dk/en/analyses-and-statistics/technology-data-generation-electricity-and-district-heating

[18] The Future of Geothermal Energy – Analysis.
URL https://www.iea.org/reports/the-future-of-geothermal-energy

[19] Projected Costs of Generating Electricity.
URL https://www.oecd-nea.org/jcms/pl_28612/projected-costs-of-generating-electricity

[20] Technology Data for Energy Storage | Energistyrelsen.
URL https://ens.dk/en/analyses-and-statistics/technology-data-energy-storage

[21] UCloud User Guide — UCloud.
URL https://docs.cloud.sdu.dk/index.html

[22] Barrier Logging - Gurobi Optimizer Reference Manual.
URL https://docs.gurobi.com/projects/optimizer/en/current/concepts/logging/barrier.html

[23] Energy Balance - Documentation.
URL https://docs.pypsa.org/latest/user-guide/optimization/energy-balance/

[24] T. T. Pedersen, M. Victoria, M. G. Rasmussen, G. B. Andresen, Modeling all alternative solutions for highly renewable energy systems 234 121294. doi:10.1016/j.energy.2021.121294.
URL https://www.sciencedirect.com/science/article/pii/S0360544221015425

[25] A. Kies, Deriving the zero-profit condition for pypsa, unpublished internal document.

## 9. Appendices

### *9.1. Appendix A: Generator cost assumptions*

Table 7: Generator technology cost assumptions used in the optimization model.

| **Technology** | **Capital Cost (EUR/MW/a)** | **Marginal Cost (EUR/MWh)** |
|---|---|---|
| CCGT | 99.0k | 47.20 |
| OCGT | 45.1k | 58.38 |
| biomass | 379.2k | 14.96 |
| coal | 180.2k | 24.10 |
| geothermal | 490.7k | 0.00 |
| lignite | 180.2k | 13.49 |
| nuclear | 535.5k | 16.90 |
| offwind-ac | 196.7k | 0.01 |
| offwind-dc | 219.4k | 0.01 |
| oil | 37.9k | 130.23 |
| onwind | 133.8k | 0.01 |
| ror | 270.9k | 0.00 |
| solar | 55.1k | 0.01 |

*9.2. Appendix B: Storage cost assumptions*

Table 8: Storage technology cost assumptions extracted from the optimized network.

| **Technology** | **Capital Cost (EUR/MW/a)** | **Marginal Cost (EUR/MWh)** |
|---|---|---|
| Battery | 13.7k | 0.00 |
| Hydrogen | 205.0k | 0.00 |
| PHS | 160.6k | 0.00 |
| Hydro | 0.0k | 0.00 |

Table 9: Installed generation capacity across decarbonization levels. Mean capacity (GW) with coefficient of variation (%) in parentheses. Cell shading shows K-sensitivity: green = capacity increases with K (favored by decentralization), red = capacity decreases with K (disfavored by decentralization), unshaded = insensitive to K-constraint.

| Technology | 0% | 50% | 80% | 100% |
|---|---|---|---|---|
| Solar PV | 424.3 (0%) | 564.7 (11%) | 666.2 (4%) | 1166.1 (23%) |
| Offshore Wind (DC) | 229.8 (0%) | 414.2 (2%) | 510.4 (1%) | 562.5 (15%) |
| Offshore Wind (AC) | 76.1 (0%) | 111.5 (14%) | 103.3 (14%) | 378.8 (53%) |
| Onshore Wind | 108.5 (0%) | 122.3 (5%) | 128.0 (7%) | 182.7 (24%) |
| CCGT (Gas) | 156.0 (0%) | 156.0 (0%) | 156.0 (0%) | 156.0 (0%) |
| Nuclear | 130.4 (0%) | 130.4 (0%) | 130.4 (0%) | 130.4 (0%) |
| Coal | 119.6 (0%) | 119.6 (0%) | 119.6 (0%) | 119.6 (0%) |
| Lignite | 50.2 (0%) | 50.2 (0%) | 50.2 (0%) | 50.2 (0%) |
| OCGT (Gas) | 39.7 (0%) | 39.7 (0%) | 39.7 (0%) | 39.7 (0%) |
| Run-of-River Hydro | 34.5 (0%) | 34.5 (0%) | 34.5 (0%) | 34.5 (0%) |
| Oil | 32.0 (0%) | 32.0 (0%) | 32.0 (0%) | 32.0 (0%) |
| Biomass | 9.9 (0%) | 9.9 (0%) | 9.9 (0%) | 9.9 (0%) |

### 9.4. *Appendix D: Annual generation evolution for decentralization and decarbonization*

The total annual generation from the four freely expandable renewable sources, across the different levels of decentralization and decarbonization can be seen below in Figure 31. In order to de-clutter the visualization each technology is represented by its k=1 scenario and the mean across the remaining K-constraints.

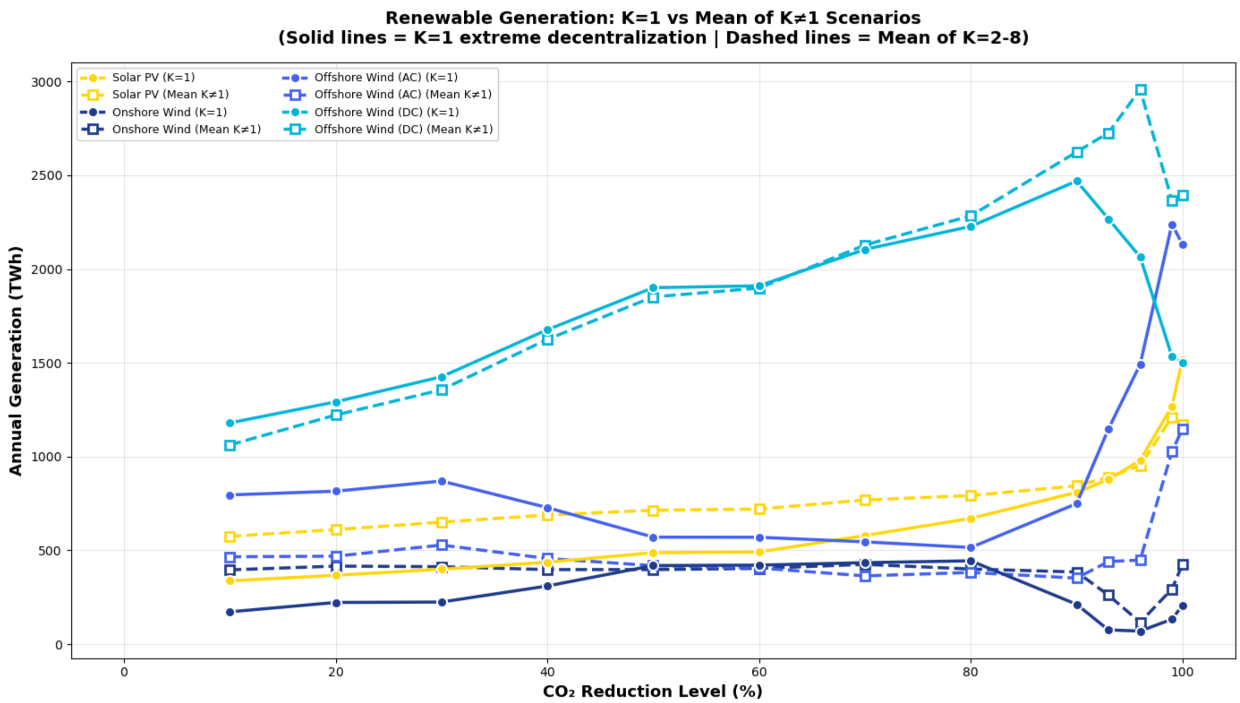


Figure 31: Total annual generation from renewable sources across K-constraints and decarbonization levels .

### *9.5. Appendix E: Coefficient of Variance analysis for Storage capacity*

Table 10: Sensitivity of Storage Capacity to K-Constraint Levels

| Storage Technology | Avg Capacity (GW) | Avg CV (%) | Min CV (%) | Max CV (%) | Avg Range (%) |
|---|---|---|---|---|---|
| Battery | 686.7 | 5.50 | 0.00 | 20.02 | 18.13 |
| Hydro Storage | 99.6 | 0.00 | 0.00 | 0.00 | 0.00 |
| Hydrogen | 0.0 | 142.38 | 0.00 | 264.14 | 438.71 |
| Pumped Hydro | 54.6 | 0.00 | 0.00 | 0.00 | 0.00 |

*Note:* CV = Coefficient of Variation. Range = (Max - Min) / Mean $\times$ 100%. Statistics calculated across all K-constraint levels (K=1 through K=$\infty$).

## *9.6. Appendix F: Mean capacity factors*

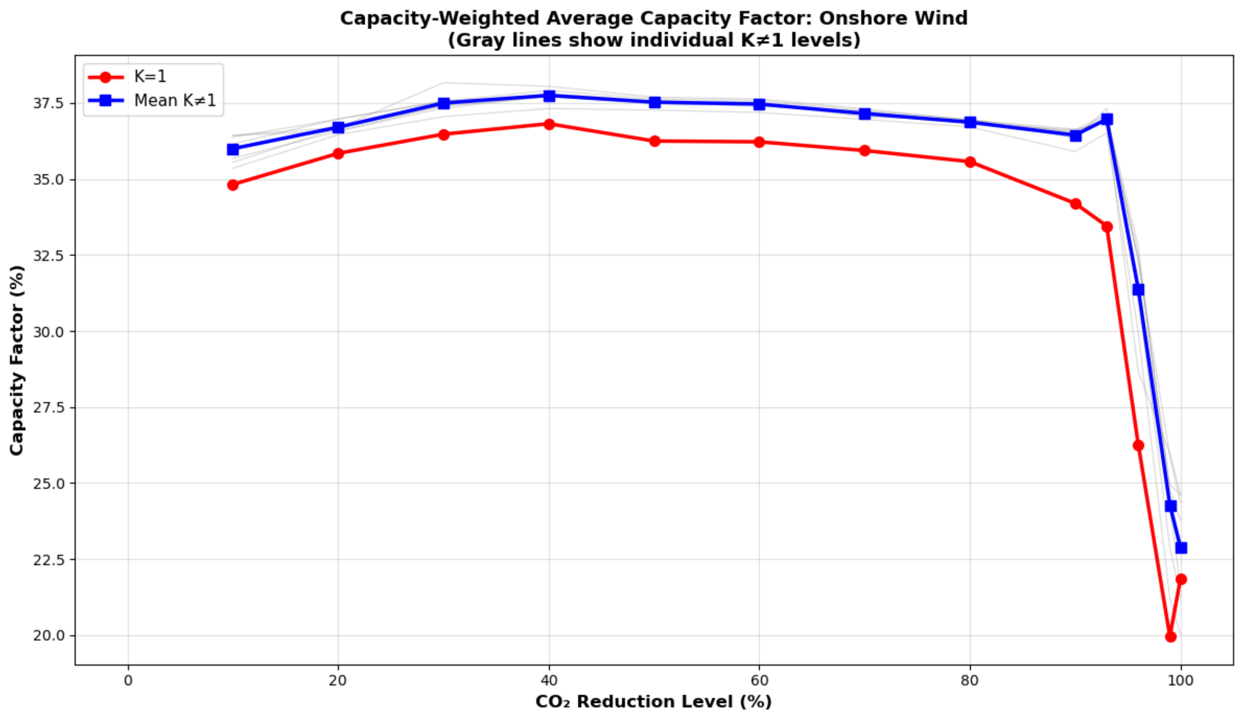


Figure 32: Mean capacity factor for onshore wind across K-constraints and decarbonization levels.

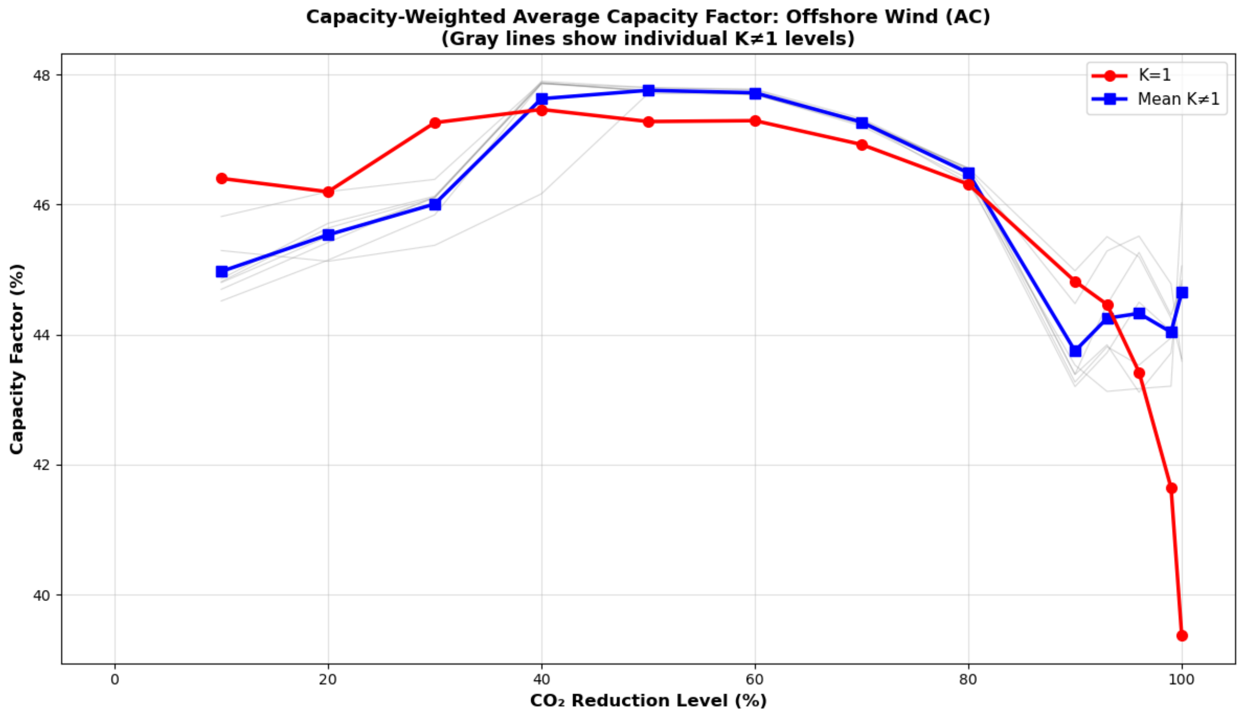


Figure 33: Mean capacity factor for offshore wind (AC) across K-constraints and decarbonization levels.

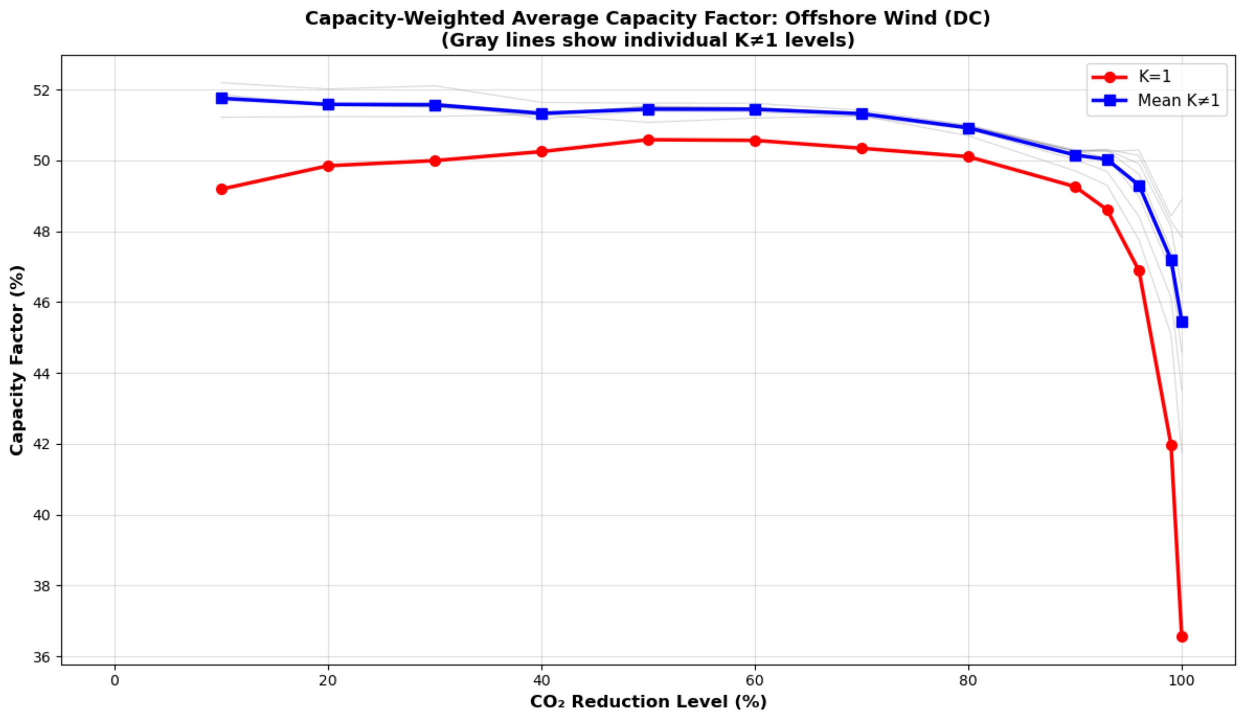


Figure 34: Mean capacity factor for offshore wind (DC) across K-constraints and decarbonization levels.

*9.7. Appendix G: Cost-Equity tradeoff*

Table 11: Cost-Equity Trade-off Across All K-Constraint Levels (Part 1: 10-40% Decarbonization)

| **Decarb Level** | **K Value** | **Cost Premium (%)** | **Equity Benefit (%)** | **Cost (% of K=1)** | **Equity (% of K=1)** | **Efficiency vs K=1** | **Efficiency vs K=$\infty$** |
|---|---|---|---|---|---|---|---|
| 10% | K=1 | 3.32 | 98.5 | 100.0 | 100.0 | 1.0x | 29.7x |
| | K=2 | 1.10 | 91.4 | 33.2 | 92.7 | 2.8x | 82.8x |
| | K=3 | 0.66 | 85.9 | 19.8 | 87.2 | 4.4x | 130.8x |
| | K=4 | 0.52 | 82.4 | 15.7 | 83.6 | 5.3x | 158.4x |
| | K=5 | 0.47 | 79.9 | 14.1 | 81.1 | 5.8x | 170.9x |
| | K=6 | 0.43 | 77.7 | 12.8 | 78.9 | 6.1x | 182.1x |
| | K=7 | 0.39 | 75.7 | 11.8 | 76.8 | 6.5x | 193.2x |
| 20% | K=1 | 3.09 | 98.4 | 100.0 | 100.0 | 1.0x | 31.9x |
| | K=2 | 1.02 | 89.9 | 33.0 | 91.3 | 2.8x | 88.3x |
| | K=3 | 0.60 | 84.3 | 19.3 | 85.6 | 4.4x | 141.5x |
| | K=4 | 0.45 | 80.5 | 14.6 | 81.8 | 5.6x | 178.6x |
| | K=5 | 0.38 | 77.3 | 12.4 | 78.5 | 6.3x | 202.4x |
| | K=6 | 0.34 | 74.9 | 11.0 | 76.1 | 6.9x | 221.1x |
| | K=7 | 0.31 | 72.3 | 9.9 | 73.4 | 7.4x | 236.7x |
| 30% | K=1 | 2.82 | 98.3 | 100.0 | 100.0 | 1.0x | 34.8x |
| | K=2 | 0.83 | 88.3 | 29.3 | 89.8 | 3.1x | 106.7x |
| | K=3 | 0.48 | 83.0 | 17.0 | 84.4 | 5.0x | 173.3x |
| | K=4 | 0.36 | 78.3 | 12.8 | 79.7 | 6.2x | 216.6x |
| | K=5 | 0.30 | 74.7 | 10.7 | 76.0 | 7.1x | 246.1x |
| | K=6 | 0.26 | 71.6 | 9.3 | 72.8 | 7.8x | 273.2x |
| | K=7 | 0.23 | 68.3 | 8.2 | 69.5 | 8.5x | 296.0x |
| 40% | K=1 | 2.59 | 98.0 | 100.0 | 100.0 | 1.0x | 37.8x |
| | K=2 | 0.60 | 85.3 | 23.1 | 87.0 | 3.8x | 142.2x |
| | K=3 | 0.31 | 77.9 | 11.8 | 79.5 | 6.7x | 253.9x |
| | K=4 | 0.23 | 73.5 | 8.8 | 75.0 | 8.5x | 322.4x |
| | K=5 | 0.19 | 69.6 | 7.2 | 71.0 | 9.9x | 373.6x |
| | K=6 | 0.16 | 65.4 | 6.1 | 66.7 | 11.0x | 414.4x |
| | K=7 | 0.14 | 61.6 | 5.2 | 62.8 | 12.1x | 456.0x |

*Note:* Cost premium and equity benefit calculated relative to K=$\infty$. 'Cost (% of K=1)' and 'Equity (% of K=1)' show each K-value's cost penalty and equity benefit as percentage of K=1's penalty/benefit. 'Efficiency vs K=1' = (Equity % of K=1) / (Cost % of K=1). 'Efficiency vs K=$\infty$' = (Equity Benefit %) / (Cost Premium %), measuring absolute decentralization effectiveness per unit cost.

Table 12: Cost-Equity Trade-off Across All K-Constraint Levels (Part 2: 50-80% Decarbonization)

| **Decarb Level** | **K Value** | **Cost Premium (%)** | **Equity Benefit (%)** | **Cost (% of K=1)** | **Equity (% of K=1)** | **Efficiency vs K=1** | **Efficiency vs K=∞** |
|---|---|---|---|---|---|---|---|
| 50% | K=1 | 2.51 | 98.1 | 100.0 | 100.0 | 1.0x | 39.1x |
| | K=2 | 0.49 | 85.0 | 19.6 | 86.7 | 4.4x | 173.4x |
| | K=3 | 0.22 | 78.1 | 9.0 | 79.7 | 8.9x | 347.8x |
| | K=4 | 0.15 | 72.5 | 6.1 | 73.9 | 12.0x | 470.9x |
| | K=5 | 0.12 | 69.1 | 4.9 | 70.4 | 14.4x | 561.3x |
| | K=6 | 0.10 | 65.6 | 4.1 | 66.9 | 16.3x | 637.5x |
| | K=7 | 0.09 | 61.8 | 3.4 | 63.0 | 18.3x | 716.0x |
| 60% | K=1 | 2.44 | 98.1 | 100.0 | 100.0 | 1.0x | 40.3x |
| | K=2 | 0.46 | 85.5 | 18.9 | 87.1 | 4.6x | 186.0x |
| | K=3 | 0.21 | 78.9 | 8.8 | 80.4 | 9.1x | 367.8x |
| | K=4 | 0.15 | 73.2 | 6.0 | 74.6 | 12.4x | 498.8x |
| | K=5 | 0.12 | 69.9 | 4.8 | 71.3 | 14.8x | 597.3x |
| | K=6 | 0.10 | 66.5 | 4.0 | 67.8 | 17.0x | 682.8x |
| | K=7 | 0.08 | 62.8 | 3.3 | 64.0 | 19.1x | 771.0x |
| 70% | K=1 | 2.36 | 97.9 | 100.0 | 100.0 | 1.0x | 41.5x |
| | K=2 | 0.40 | 83.1 | 16.8 | 84.9 | 5.1x | 210.2x |
| | K=3 | 0.19 | 75.3 | 8.0 | 77.0 | 9.6x | 398.9x |
| | K=4 | 0.13 | 68.5 | 5.5 | 70.0 | 12.7x | 528.6x |
| | K=5 | 0.10 | 63.4 | 4.2 | 64.7 | 15.6x | 646.8x |
| | K=6 | 0.08 | 58.9 | 3.4 | 60.2 | 17.8x | 739.0x |
| | K=7 | 0.07 | 54.5 | 2.8 | 55.7 | 20.1x | 836.7x |
| 80% | K=1 | 2.32 | 97.6 | 100.0 | 100.0 | 1.0x | 42.1x |
| | K=2 | 0.33 | 79.7 | 14.4 | 81.7 | 5.7x | 238.9x |
| | K=3 | 0.15 | 70.7 | 6.3 | 72.4 | 11.6x | 487.0x |
| | K=4 | 0.09 | 63.0 | 4.0 | 64.5 | 16.1x | 676.9x |
| | K=5 | 0.07 | 57.5 | 3.1 | 58.9 | 19.3x | 810.3x |
| | K=6 | 0.06 | 52.5 | 2.5 | 53.8 | 21.9x | 922.2x |
| | K=7 | 0.05 | 47.0 | 2.0 | 48.1 | 24.5x | 1029.3x |

*Note:* Cost premium and equity benefit calculated relative to K=∞. 'Cost (% of K=1)' and 'Equity (% of K=1)' show each K-value's cost penalty and equity benefit as percentage of K=1's penalty/benefit. 'Efficiency vs K=1' = (Equity % of K=1) / (Cost % of K=1). 'Efficiency vs K=∞' = (Equity Benefit %) / (Cost Premium %), measuring absolute decentralization effectiveness per unit cost.

Table 13: Cost-Equity Trade-off Across All K-Constraint Levels (Part 3: 90-99% Decarbonization)

| **Decarb Level** | **K Value** | **Cost Premium (%)** | **Equity Benefit (%)** | **Cost (% of K=1)** | **Equity (% of K=1)** | **Efficiency vs K=1** | **Efficiency vs K=$\infty$** |
|---|---|---|---|---|---|---|---|
| 90% | K=1 | 2.77 | 98.3 | 100.0 | 100.0 | 1.0x | 35.5x |
| | K=2 | 0.39 | 84.3 | 14.1 | 85.8 | 6.1x | 215.9x |
| | K=3 | 0.17 | 75.9 | 6.3 | 77.2 | 12.4x | 438.9x |
| | K=4 | 0.09 | 68.2 | 3.4 | 69.4 | 20.4x | 724.5x |
| | K=5 | 0.07 | 62.1 | 2.5 | 63.2 | 25.4x | 903.4x |
| | K=6 | 0.06 | 58.4 | 2.1 | 59.4 | 28.0x | 996.5x |
| | K=7 | 0.05 | 54.6 | 1.8 | 55.5 | 30.8x | 1093.2x |
| 93% | K=1 | 3.64 | 98.5 | 100.0 | 100.0 | 1.0x | 27.1x |
| | K=2 | 0.66 | 86.3 | 18.1 | 87.5 | 4.8x | 131.0x |
| | K=3 | 0.30 | 78.3 | 8.3 | 79.5 | 9.5x | 257.6x |
| | K=4 | 0.15 | 72.6 | 4.2 | 73.7 | 17.4x | 470.4x |
| | K=5 | 0.10 | 67.1 | 2.7 | 68.1 | 25.5x | 690.1x |
| | K=6 | 0.07 | 62.8 | 2.0 | 63.7 | 31.4x | 849.5x |
| | K=7 | 0.06 | 59.7 | 1.8 | 60.6 | 34.5x | 934.3x |
| 96% | K=1 | 5.56 | 99.0 | 100.0 | 100.0 | 1.0x | 17.8x |
| | K=2 | 1.74 | 88.3 | 31.2 | 89.2 | 2.9x | 50.9x |
| | K=3 | 0.98 | 83.3 | 17.7 | 84.2 | 4.8x | 84.7x |
| | K=4 | 0.63 | 79.5 | 11.3 | 80.3 | 7.1x | 126.9x |
| | K=5 | 0.38 | 76.6 | 6.9 | 77.4 | 11.2x | 199.8x |
| | K=6 | 0.22 | 72.2 | 3.9 | 73.0 | 18.6x | 330.6x |
| | K=7 | 0.16 | 68.4 | 2.8 | 69.1 | 24.7x | 439.8x |
| 99% | K=1 | 9.90 | 99.4 | 100.0 | 100.0 | 1.0x | 10.0x |
| | K=2 | 3.98 | 91.6 | 40.2 | 92.2 | 2.3x | 23.0x |
| | K=3 | 2.30 | 87.3 | 23.2 | 87.8 | 3.8x | 38.0x |
| | K=4 | 1.49 | 84.2 | 15.1 | 84.8 | 5.6x | 56.5x |
| | K=5 | 1.15 | 81.3 | 11.6 | 81.8 | 7.1x | 70.8x |
| | K=6 | 0.94 | 78.3 | 9.5 | 78.8 | 8.3x | 83.4x |
| | K=7 | 0.84 | 75.7 | 8.5 | 76.2 | 9.0x | 90.2x |

*Note:* Cost premium and equity benefit calculated relative to K=$\infty$. 'Cost (% of K=1)' and 'Equity (% of K=1)' show each K-value's cost penalty and equity benefit as percentage of K=1's penalty/benefit. 'Efficiency vs K=1' = (Equity % of K=1) / (Cost % of K=1). 'Efficiency vs K=$\infty$' = (Equity Benefit %) / (Cost Premium %), measuring absolute decentralization effectiveness per unit cost.

Table 14: Cost-Equity Trade-off Across All K-Constraint Levels (Part 4: 100% Decarbonization)

| Decarb Level | K Value | Cost Premium (%) | Equity Benefit (%) | Cost (% of K=1) | Equity (% of K=1) | Efficiency vs K=1 | Efficiency vs K=$\infty$ |
|---|---|---|---|---|---|---|---|
| 100% | K=1 | 18.11 | 99.4 | 100.0 | 100.0 | 1.0x | 5.5x |
| | K=2 | 7.99 | 91.8 | 44.1 | 92.3 | 2.1x | 11.5x |
| | K=3 | 4.78 | 88.0 | 26.4 | 88.4 | 3.4x | 18.4x |
| | K=4 | 3.32 | 84.8 | 18.3 | 85.3 | 4.7x | 25.6x |
| | K=5 | 2.46 | 81.1 | 13.6 | 81.6 | 6.0x | 33.0x |
| | K=6 | 1.91 | 78.3 | 10.5 | 78.7 | 7.5x | 41.0x |
| | K=7 | 1.56 | 76.1 | 8.6 | 76.5 | 8.9x | 48.7x |

*Note:* Cost premium and equity benefit calculated relative to K=$\infty$. 'Cost (% of K=1)' and 'Equity (% of K=1)' show each K-value's cost penalty and equity benefit as percentage of K=1's penalty/benefit. 'Efficiency vs K=1' = (Equity % of K=1) / (Cost % of K=1). 'Efficiency vs K=$\infty$' = (Equity Benefit %) / (Cost Premium %), measuring absolute decentralization effectiveness per unit cost.

Table 15: $K = \infty$ (No Constraint): Renewable Capacity per Load Distribution Statistics at 100% Decarbonization

| **Metric** | **Value** |
|---|---|
| Total Nodes (with load) | 37 |
| Mean Ratio ( $\mathrm{MW_{renewable}}/\mathrm{MW_{load}}$) | 9.29 |
| Standard Deviation (SDRCL, $\sigma$) | 28.96 |
| Upper Bound ($\mu + \sigma$) | 38.25 |
| Lower Bound ($\mu - \sigma$) | -19.67 |
| *Distribution:* | |
| Nodes Above $\mu + \sigma$ | 2 |
| Nodes Within $\pm\sigma$ | 35 |
| Nodes Below $\mu - \sigma$ | 0 |

Table 16: $K = 7$: Renewable Capacity per Load Distribution Statistics at 100% Decarbonization

| **Metric** | **Value** |
|---|---|
| Total Nodes (with load) | 37 |
| Mean Ratio ( $\mathrm{MW_{renewable}}/\mathrm{MW_{load}}$) | 4.25 |
| Standard Deviation (SDRCL, $\sigma$) | 6.93 |
| Upper Bound ($\mu + \sigma$) | 11.18 |
| Lower Bound ($\mu - \sigma$) | -2.68 |
| *Distribution:* | |
| Nodes Above $\mu + \sigma$ | 6 |
| Nodes Within $\pm\sigma$ | 31 |
| Nodes Below $\mu - \sigma$ | 0 |

Table 17: $K = 1$: Renewable Capacity per Load Distribution Statistics at 100% Decarbonization

| **Metric** | **Value** |
|---|---|
| Total Nodes (with load) | 37 |
| Mean Ratio ( $MW_{renewable}/MW_{load}$) | 4.25 |
| Standard Deviation (SDRCL, $\sigma$) | 0.16 |
| Upper Bound ($\mu + \sigma$) | 4.41 |
| Lower Bound ($\mu - \sigma$) | 4.09 |
| *Distribution:* | |
| Nodes Above $\mu + \sigma$ | 2 |
| Nodes Within $\pm\sigma$ | 35 |
| Nodes Below $\mu - \sigma$ | 0 |

### *9.8. Appendix H: Distribution statistics*

### *9.9. Appendix I: Figures for decentralization evolution*

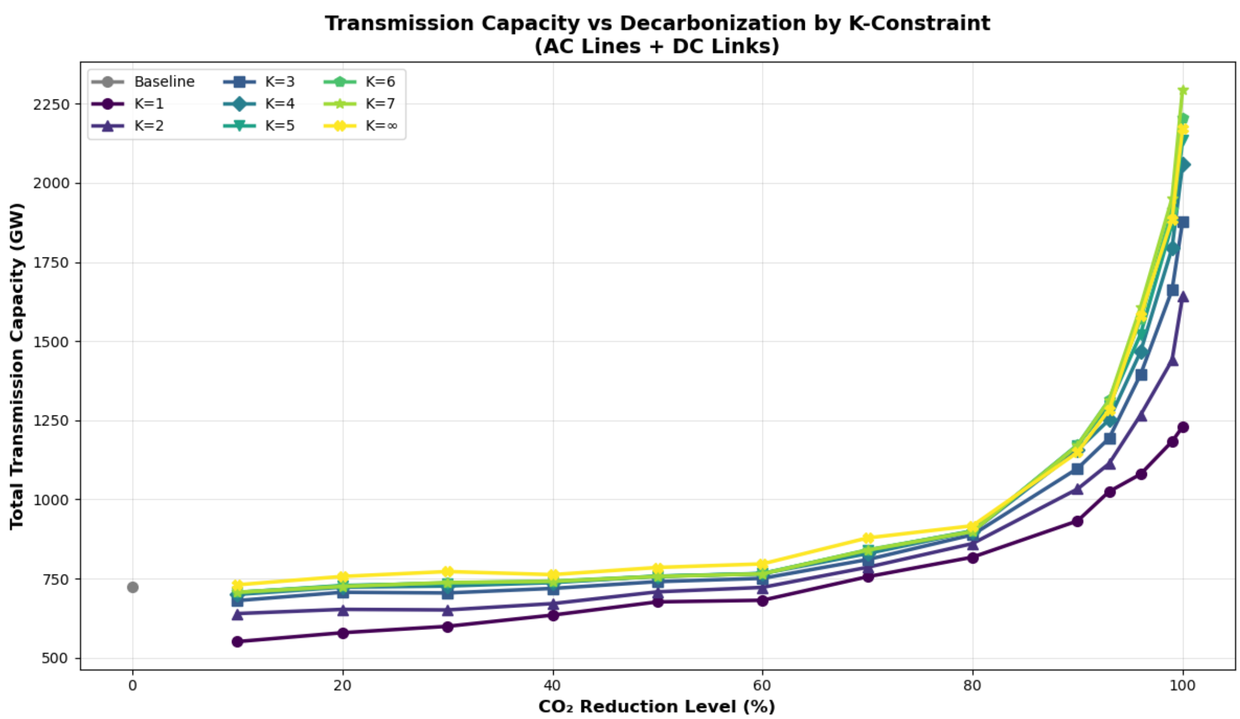


Figure 35: Total installed transmission capacity across K-constraints and decarbonization levels.

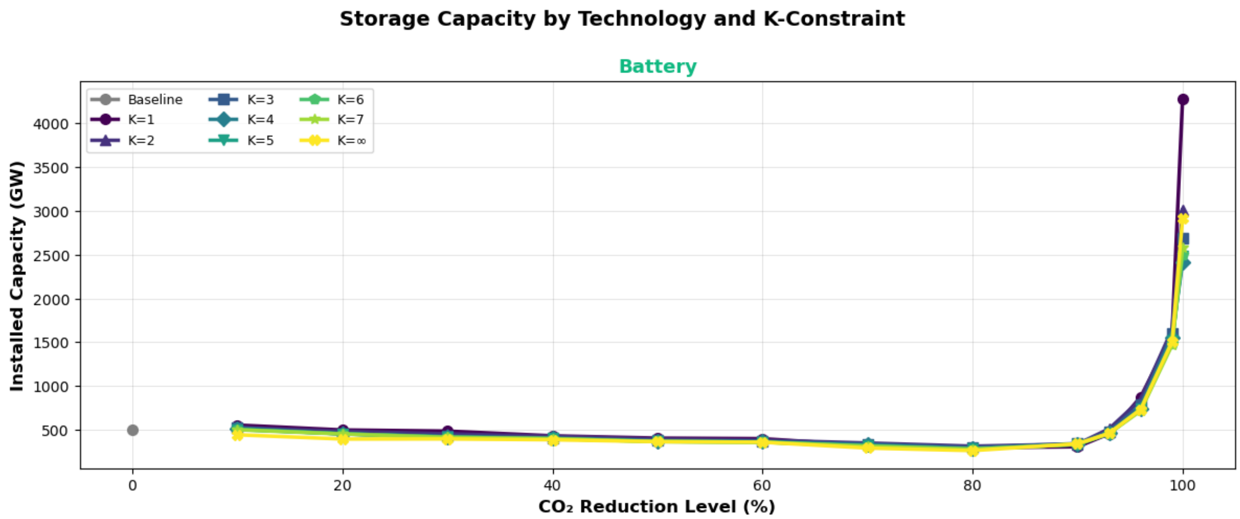


Figure 36: Installed battery and hydrogen storage capacity across K-constraints and decarbonization levels.

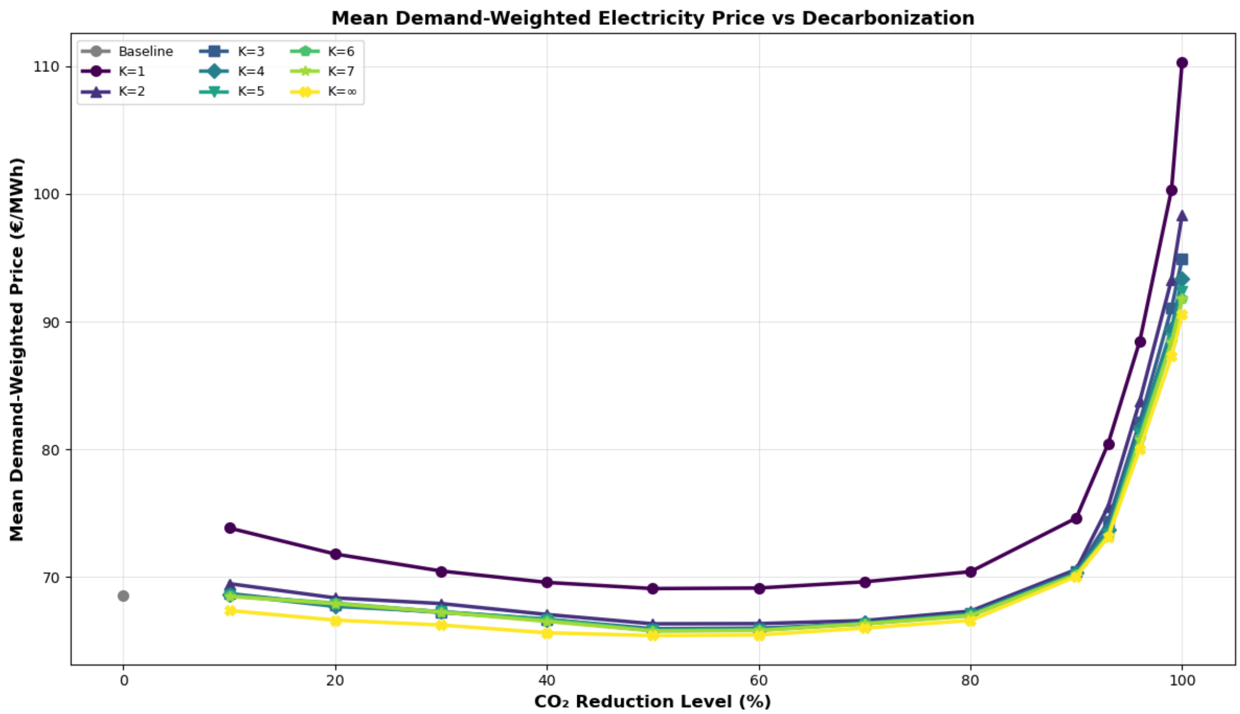


Figure 37: Mean marginal demand-weighted price across K-constraints and decarbonization levels.

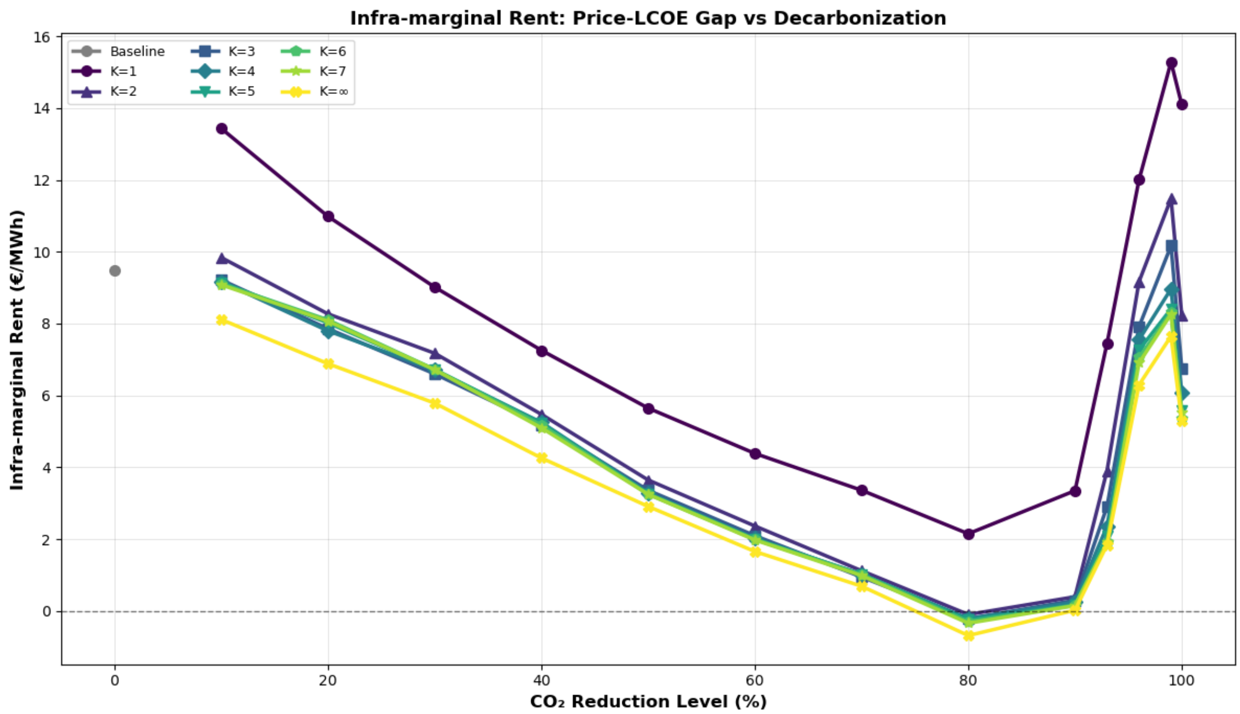


Figure 38: Total scarcity rent across K-constraints and decarbonization levels.